EXAM PACK

EEE443

Power Plant Engineering



Exam Pack

Session 2020–21 *Solutions and formula sheet of the 4th Year 1st Semester Final Examination, 2024*

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Prepared by Nur Mohammad

October 2026

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PAST PAPER SOLUTIONS

EEE443

Power Plant Engineering

Question Paper Solutions

Session 2020–21

4th Year 1st Semester Final Examination, 2024

QUESTION MAP

Q1	Q2	Q3	Q4	Q5	Q6
25 marks	25 marks	25 marks	25 marks	25 marks	25 marks

EACH ANSWER

Theory → Answer → Keyword Explanation → Notes and Remarks → Final Concise Answer

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October 2026

BACK OF THE COVER

The paper at a glance

3 h	100	4 of 6	45 min
Total time	Total marks	Questions to answer	Per question

Time = marks × 1.8 min (180 min for 100 marks). Tick the questions you will answer; each sub-part links to its solution.

Q	Sub-part	Marks	Time	Pick
PART – A · (ANSWER TWO OF THE FOLLOWING QUESTIONS)				
Q1	(a) Factors that Improve a Thermal Power Station	5	9 min	☐
	(b) Characteristics of Fuels	5	9 min	
	(c) Combined Cycle Power Generation	5	9 min	
	(d) Base Load and Peak Load Plants, and the Load Factors	10	18 min	
Q2	(a) Importance of Load Factor and Diversity Factor	3	5 min	☐
	(b) Variable Load on a Power Station	4	7 min	
	(c) Economic Load Sharing: Equal Incremental Heat Rates	10	18 min	
	(d) Economic Loading of Two Units A and B	8	14 min	
Q3	(a) Steam Generating Part of a Coal-Fired Power Plant	7	13 min	☐
	(b) Water-Tube and Fire-Tube Boilers	5	9 min	
	(c) How a Superheater and an Economiser Improve a Boiler	5	9 min	
	(d) Efficiency of the Boiler, the Economiser and the Whole Plant	8	14 min	
PART – B · (ANSWER TWO OF THE FOLLOWING QUESTIONS)				
Q4	(a) Plant Use Factor and Plant Utility Factor	4	7 min	☐
	(b) Types of Nuclear Reactors	5	9 min	
	(c) Daily Energy, Reserve and Maximum Energy of a Station	8	14 min	
	(d) Kaplan and Francis Turbines	8	14 min	
Q5	(a) Flat Demand Rate and Straight Line Meter Rate	5	9 min	☐
	(b) Generating Cost per kWh	5	9 min	
	(c) Spillways, Surge Tanks and Draft Tubes	6	11 min	
	(d) Two-Part Tariff of a Supply Company	9	16 min	
Q6	(a) Methods of Determining Depreciation	7	13 min	☐
	(b) Pulverisation and the Ball Mill	6	11 min	
	(c) Impulse and Reaction Turbines	6	11 min	
	(d) Advantages and Disadvantages of a Hydro-Electric Power Plant	6	11 min	

Formulas follow V.K. Mehta, Principles of Power System (load curves and factors, economics, tariffs, hydro and nuclear plants) and P.K. Nag, Power Plant Engineering (fuels, boilers, economisers, turbines).

Shahjalal University of Science and Technology

Department of Electrical and Electronic Engineering
4th Year 1st Semester Final Examination, 2024
Course Code: EEE 443, Session: 2020-21
Course Title: Power Plant Engineering

Full Marks: 100 **Credit: 3.0** **Exam Duration: 3 Hours**

Part – A

*(Answer **two** of the following questions)*

1. (a) List the factors that contribute to improving the generating capacity of a thermal power station. 5

   (b) Briefly discuss the characteristics of fuels. 5

   (c) What is meant by combined cycle power generation? 5

   (d) What are the requirements of a base load and peak load power plant? The annual peak load on a 30 MW power station is 25 MW. The power station supplies loads having a maximum demands of 10MW, 8.5 MW, 5 MW and 4.5 MW. The annual load factor is 45%. Find:

   (i) Average load,
   (ii) Energy supplied every year,
   (iii) Diversity factor,
   (iv) Demand/factor. [*sic*: should be "Demand factor"] 10

2. (a) What is the importance of load factor and diversity factor? 3

   (b) Why is the load on a power station variable? What are the effects of variable load on the operation of the power station? 4

   (c) Show that for the economic load sharing, the incremental heat rates are equal. 10

   (d) The incremental fuel costs for two generating units A and B of a power plant are given by the following equations: $\frac{dF_A}{dP_A} = 0.06P_A + 11.4$, $\frac{dF_B}{dP_B} = 0.07P_B + 10$, where P is in mega-watts and F is in takas per hour.

   (i) Find the economic loading of the two units when the total load to be supplied by the power station is 150 MW.
   (ii) Find the loss in fuel cost per hour if the load is equally shared by the two units. 8

3. (a) Draw the schematics of steam generating part of a coal fire power plant showing all basic components and direction of steam flow. 7

   (b) Differentiate between water tube boiler and fire tube boiler. 5

   (c) A superheater and an economizer improve the performance of a boiler-How? 5

   (d) The following data is supplied for a boiler plant: Boiler: Mass of coal supplied = 220 kg /hr. Calorific value of coal = 7200 kcal /kg. Mass of feed water = 200 kg/hr [*sic*: should be 2000 kg/hr]. Enthalpy of steam produced = 670 kcal/kg. Economizer: Inlet temperature of feed water = 18°C. Outlet temperature of water = 88°C. Atmospheric temperature = 19°C. Temperature of flue gas entering = 350°C. Mass of flue gases = 4150 kg/hr. Determine the following

   (i) Efficiency of boiler.
   (ii) Efficiency of economizer.

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(iii) Efficiency of whole boiler plant. 8

Part – B

*(Answer **two** of the following questions)*

4. (a) Define plant use factor and plant utility factor. 4

(b) Mention the types of reactors used in nuclear power plants. 5

(c) A generating station has a maximum demand of 20 MW, a load factor of 60%, a plant capacity factor of 48% and a plant use factor of 80%. Find:

(i) The daily energy produced
(ii) The reserve capacity of the plant
(iii) The maximum energy that could be produced daily if the plant was running all the time
(iv) The maximum energy that could be produced daily if the plant was running fully loaded and operating as per schedule. 8

(d) With diagram, write short notes on Kaplan and Francis turbine. 8

5. (a) Write short notes on the following type of tariffs:

(i) Flat demand rate
(ii) Straight line meter rate 5

(b) Estimate the generating cost per KWh delivered from a generating station from the following data: Plant capacity = 50 MW. Annual load factor = 40%. Capital cost = 1.2 crores. Annual cost of wages, taxation, etc. = Taka 4 lakhs. Cost of fuel, lubrication, maintenance, etc. = 1.0 paisa/kWh generated. Interest 5% per annum, depreciation 6% per annum of initial value. 5

(c) Write short note on-

(i) Spillways
(ii) Surge tanks
(iii) Draft tubes 6

(d) An electric supply company having a maximum load of 50 MW generates 18×10^7 units per annum and the supply consumers have an aggregate demand of 75 MW. The annual expenses including capital charges are: For fuel = Taka 90 lakhs. Fixed charges concerning generation = Taka 28 lakhs. Fixed charges concerning transmission and distribution = Taka 32 lakhs. Assuming 90% of the fuel cost is essential to running charges and the loss in transmission and distribution as 15% of KWh generated, deduce a two part tariff to find the actual cost of supply to the consumers. 9

6. (a) What are the methods of determining depreciation cost in power plants? Briefly explain one of the methods. 7

(b) Why is pulverization done in a coal fired power plant? Give description of any basic type of pulverizers. 6

(c) Write short notes on:

(i) Impulse turbine
(ii) Reaction turbines 6

(d) Discuss the advantages and disadvantages of a Hydro-electric power plant. 6

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1. (a) List the factors that contribute to improving the generating capacity of a thermal power station. Marks: 5
   (b) Briefly discuss the characteristics of fuels. Marks: 5
   (c) What is meant by combined cycle power generation? Marks: 5
   (d) What are the requirements of a base load and peak load power plant? The annual peak load on a 30 MW power station is 25 MW. The power station supplies loads having a maximum demands of 10MW, 8.5 MW, 5 MW and 4.5 MW. The annual load factor is 45%. Find:
      (i) Average load,
      (ii) Energy supplied every year,
      (iii) Diversity factor,
      (iv) Demand/factor. [*sic*: should be "Demand factor"] Marks: 10

1.(a) List the factors that contribute to improving the generating capacity of a thermal power station. Marks: 5

1. Theory

1. What is a thermal power station? (Core Idea)

A **thermal power station** turns the heat of burning fuel (usually coal, gas or oil) into electricity. It does this in four steps, which go round in a closed loop of water:

1. **Boiler:** the fuel burns in a furnace, and its heat boils water into high-pressure **steam**.
2. **Turbine:** the steam pushes on the blades of a turbine and makes its shaft spin.
3. **Generator:** the turbine shaft turns a generator, which produces electricity.
4. **Condenser:** the used steam is cooled back into water, and a **feed pump** sends the water back to the boiler.

This loop is called the **Rankine cycle**. The figure below shows it.

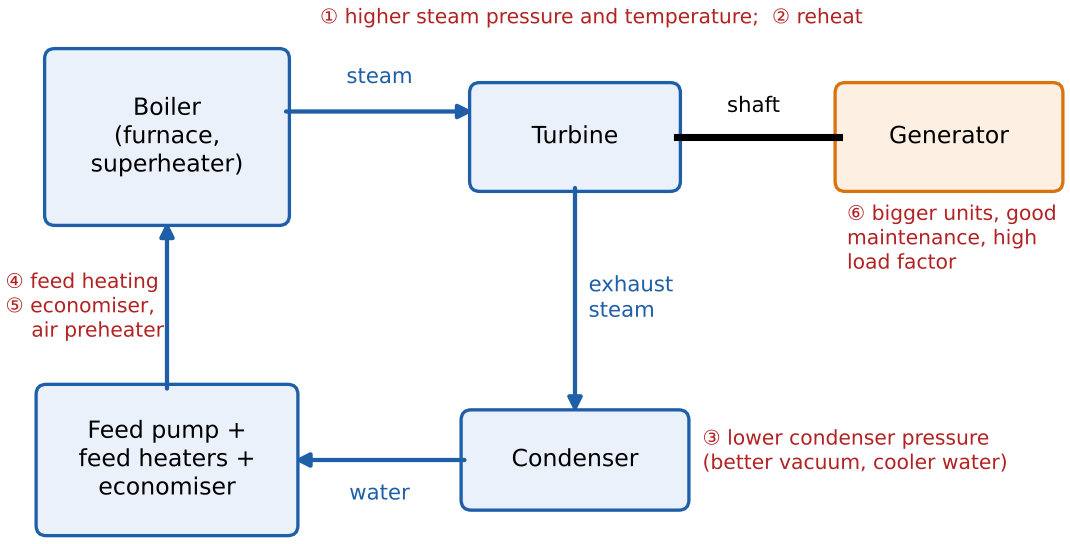


Figure 1: The steam loop of a thermal power station (Rankine cycle) with the place where each improvement acts (circled numbers)

Reading the figure. Follow the arrows round the loop: steam leaves the boiler, spins the turbine, is turned back into water in the condenser, and is pumped back to the boiler. The generator sits on the

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same shaft as the turbine. Each circled number (1 to 6) marks one of the improvements listed in the answer, placed where it acts.

2. What does "generating capacity" mean here?

The **generating capacity** of a station is how much electrical power it can deliver, in MW. For a given amount of fuel, a station delivers more power when less of the heat is wasted. So "improving the generating capacity" in practice means two things:

- getting **more electrical energy out of each kilogram of fuel** (higher **thermal efficiency**), and
- keeping the plant **available** to run near its full rating for more of the time.

3. Where does the heat go?

Only about 35–40 % of the fuel's heat becomes electricity in a steam station. The biggest losses are:

- **Heat thrown away in the condenser** (about half the fuel's heat): the steam must give up its latent heat to the cooling water before it can be pumped back.
- **Heat lost up the chimney** with the hot flue gas.
- Smaller losses: unburnt fuel, radiation from hot surfaces, friction, and the power the plant uses for its own pumps and fans.

The efficiency of any heat engine rises when it takes in heat at a **higher temperature** and rejects it at a **lower temperature**. Most of the improvements below do one of these two things, or they recover heat that would otherwise escape.

2. Answer

The generating capacity (output per unit of fuel and the power the station can reliably deliver) of a thermal power station is improved by the following:

1. **Higher steam pressure and temperature:** steam at higher pressure and temperature at the turbine inlet carries more energy per kilogram, so the cycle efficiency and the output of the same turbine increase.
2. **Reheating:** after the steam has partly expanded in the high-pressure turbine, it is sent back to the boiler and heated again before the low-pressure turbine. This raises the average temperature of heat input and keeps the last stages free of water droplets.
3. **Lower condenser pressure (better vacuum):** a lower back pressure lets the steam expand further in the turbine and do more work. Cool cooling water and a tight, air-free condenser give this.
4. **Regenerative feed-water heating:** steam bled from the turbine heats the feed water before it enters the boiler, so less fuel is needed to raise the water to boiling.
5. **Economiser and air preheater:** they use the heat left in the flue gas to warm the feed water and the combustion air, cutting the chimney loss (3(c)).
6. **Good combustion:** pulverised coal, the right amount of excess air and good mixing burn the fuel completely, so less leaves as unburnt carbon or CO.
7. **Larger units:** big turbo-generator sets have lower losses per MW than small ones.
8. **High load factor and good maintenance:** running near full load most of the time, clean boiler tubes (no scale or soot), and fewer breakdowns keep the plant at its best efficiency and available for more hours.
9. **Combined cycle:** adding a gas turbine whose hot exhaust raises steam for the steam turbine (1(c)).

3. Keyword Explanation

1. Thermal efficiency

The fraction of the heat in the fuel that comes out as electrical energy: $\eta = \dfrac{\text{electrical output}}{\text{heat input}}$.

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2. Flue gas

The hot gases (CO_2, water vapour, nitrogen, excess air) left after the fuel burns. They pass through the boiler and leave by the chimney.

3. Back pressure

The pressure at the exhaust of the turbine, set by the condenser. The lower it is, the more work each kilogram of steam does.

1.(b) Briefly discuss the characteristics of fuels. Marks: 5

1. Theory

1. What is a fuel? (Core Idea)

A **fuel** is a substance that releases heat when it burns, that is, when its carbon and hydrogen combine with the oxygen of the air:

$$C + O_2 \rightarrow CO_2 + \text{heat}, \qquad 2H_2 + O_2 \rightarrow 2H_2O + \text{heat}$$

Power stations use three kinds:

- **Solid fuels:** coal (peat, lignite, bituminous coal, anthracite, in order of increasing age and carbon content), wood, bagasse.
- **Liquid fuels:** furnace oil, diesel, made from crude petroleum.
- **Gaseous fuels:** natural gas (mainly methane, CH_4), producer gas, blast-furnace gas.

2. How is a fuel judged?

A fuel is judged by **how much heat it gives** and by **how easily and cleanly it gives it**. Two standard tests describe a coal:

- **Proximate analysis** (simple heating test): the percentages of moisture, volatile matter, fixed carbon and ash.
- **Ultimate analysis** (chemical test): the percentages of the elements carbon, hydrogen, oxygen, nitrogen and sulphur.

The characteristics in the answer come from these tests.

2. Answer

The main characteristics of a fuel are:

1. **Calorific value:** the heat released by burning 1 kg of the fuel (kcal/kg or kJ/kg). The **higher** calorific value counts the heat of condensing the water vapour formed; the **lower** value does not. A good fuel has a high calorific value (coal about 5000–8000 kcal/kg, oil about 10 000 kcal/kg).
2. **Moisture content:** water in the fuel absorbs heat to evaporate, so it lowers the useful heat. Less moisture is better.
3. **Volatile matter:** gases given off when the fuel is heated. A moderate amount helps the fuel ignite and burn with a long flame; too much gives smoke.
4. **Fixed carbon:** the carbon left after the volatile matter is driven off; it supplies most of the heat of coal. More fixed carbon means more heat.
5. **Ash content:** the non-burning mineral matter. Ash lowers the calorific value, must be handled and disposed of, and can melt into **clinker** on the grate. Less ash is better.
6. **Sulphur content:** sulphur burns to SO_2, which corrodes the boiler's cool parts and pollutes the air. It should be low.
7. **Ignition temperature:** the lowest temperature at which the fuel starts to burn. It should be moderate: low enough to light easily, high enough to store safely.
8. **Ease of handling, storage and transport:** gaseous and liquid fuels flow through pipes and burn without ash; solid fuels need crushing, conveyors and ash handling.
9. **Cost and availability:** a fuel must be cheap and available in steady supply near the station.

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In short: gaseous fuels burn most cleanly and are easiest to control; liquid fuels have the highest calorific value per kilogram; solid fuels (coal) are the cheapest and most plentiful but bring ash and handling costs.

1.(c) What is meant by combined cycle power generation? Marks: 5

1. Theory

1. Two kinds of heat engine (Core Idea)

- A **gas turbine** burns fuel (usually natural gas) in compressed air and lets the hot gas spin a turbine directly. It starts quickly but its exhaust leaves very hot, at about 500–600 °C, so a gas turbine alone wastes much heat (efficiency about 30–38 %).
- A **steam turbine** plant uses heat to boil water; it needs heat at only a few hundred °C to make good steam.

The idea of a combined cycle is that the "waste" heat of the first engine is hot enough to run the second.

2. Why does this raise the efficiency?

The gas turbine takes in heat at a very high temperature (above 1200 °C), and the steam cycle rejects heat at a low temperature (about 40 °C in the condenser). Together, the plant works between a very hot source and a cold sink, which is what makes any heat engine efficient (1(a)). The fuel is burnt once but drives two turbines.

2. Answer

Combined cycle power generation is the generation of electricity by a gas turbine and a steam turbine working together, where the hot exhaust gas of the gas turbine is used to raise steam for the steam turbine. One fuel input drives two generators.

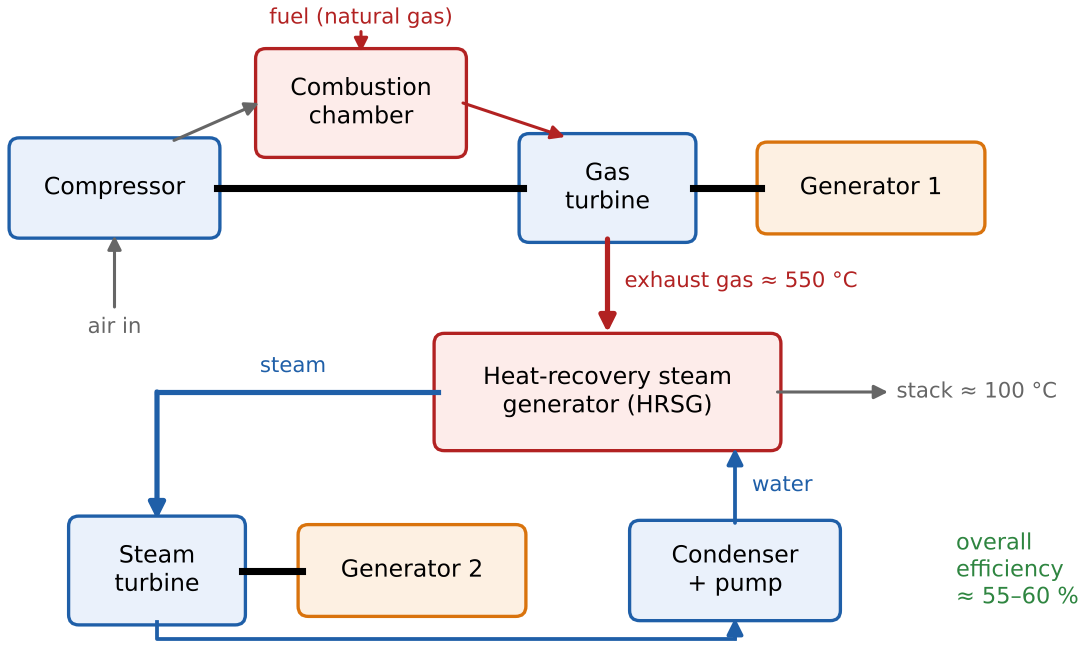


Figure 2: A combined cycle plant: the gas turbine's exhaust raises steam in a heat-recovery steam generator

Reading the figure. The top row is the gas turbine set: air is compressed, fuel is burnt in it, and the hot gas drives the gas turbine and generator 1. Its exhaust (about 550 °C) goes down to the heat-recovery steam generator, where it boils water. The steam from it drives the steam turbine and generator 2, and is condensed and pumped back.

How it works:

1. The compressor raises the pressure of air; fuel burns in this air in the combustion chamber.

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2. The hot gas expands through the gas turbine, which drives generator 1.
3. The exhaust gas, still at about 550 °C, passes through a **heat-recovery steam generator (HRSG)**, a boiler with no furnace of its own (sometimes with extra burners).
4. The steam drives a steam turbine and generator 2; the steam is condensed and returned to the HRSG.

Advantages:

- Overall efficiency of about 55–60 %, against about 35–40 % for a steam plant or a gas turbine alone.
- Less fuel and less CO_2 per kWh.
- The gas turbine part starts quickly, so the plant can follow load changes.
- Shorter construction time and less cooling water than a steam plant of the same rating.

1.(d) What are the requirements of a base load and peak load power plant? The annual peak load on a 30 MW power station is 25 MW. The power station supplies loads having a maximum demands of 10MW, 8.5 MW, 5 MW and 4.5 MW. The annual load factor is 45%. Find: (i) Average load, (ii) Energy supplied every year, (iii) Diversity factor, (iv) Demand/factor. [*sic*: should be "Demand factor"] **Marks: 10**

1. Theory

Note on the paper

The last item is printed as "Demand/factor"; it means the **demand factor**, which is what is solved here.

1. Why does the load change? (Core Idea)

The **load** on a power station is the total power its consumers draw at a moment. It changes through the day: low at night, rising in the morning, and highest in the evening, when lights, fans and cooking all run together. A graph of load against time is the **load curve**.

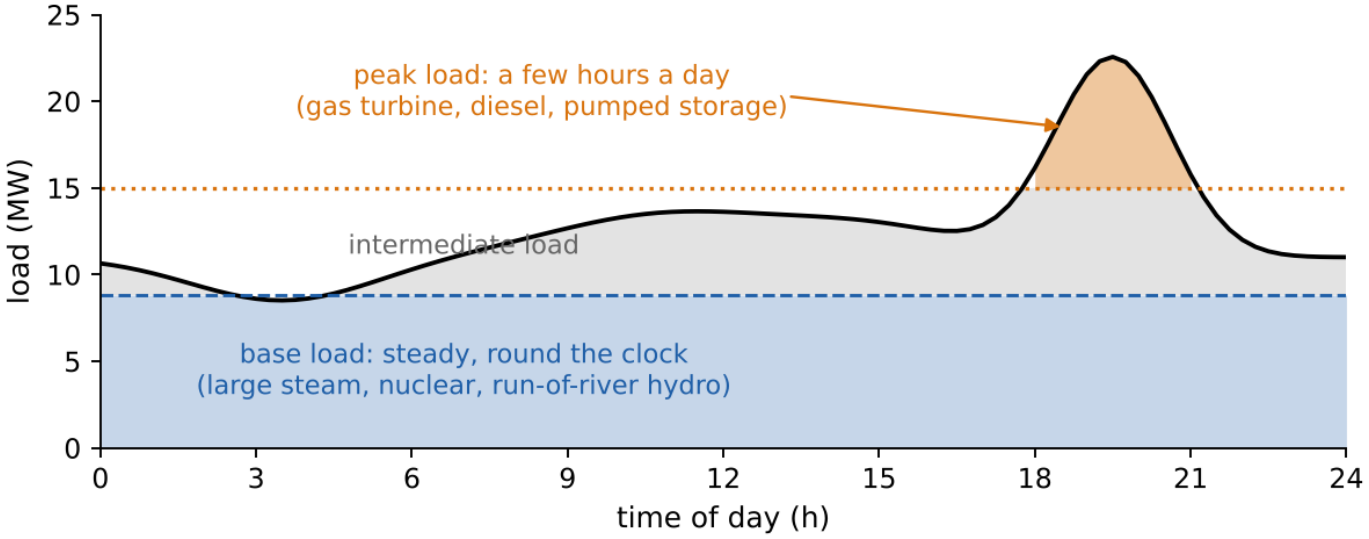


Figure 3: A daily load curve divided into base, intermediate and peak load

Reading the figure. The thick curve is the load through the day. The band below the dashed line is present for all 24 hours: that is the **base load**. The part above the dotted line appears only for a few evening hours: that is the **peak load**. The grey band between them is the intermediate load.

2. Base load and peak load plants

Since part of the load is always there and part only comes for a few hours, it pays to use two different kinds of plant:

- A **base load plant** runs all the time at nearly full output. It is worth paying a high building cost if each kWh is cheap to make.
- A **peak load plant** runs only for a few hours. Its building cost is spread over few kWh, so it must be cheap to build and quick to start, even if its fuel is dearer.

3. The numbers that describe a load

- **Maximum demand:** the highest load during the period (here the year), in MW.
- **Average load:** the energy supplied in the period divided by the number of hours: $P_{av} = \dfrac{E}{t}$.
- **Load factor:** how close the average comes to the maximum. A flat load curve has a load factor near 1:

$$\text{LF} = \frac{\text{average load}}{\text{maximum demand}}$$

- **Diversity factor:** different consumers reach their peaks at different times, so the station's peak is smaller than the sum of the consumers' peaks. The diversity factor measures this (it is always ≥ 1):

$$\text{DF} = \frac{\text{sum of the individual maximum demands}}{\text{maximum demand on the station}}$$

- **Demand factor:** the maximum demand compared with the **connected load**, the total rating of everything connected (here the 30 MW station rating). It is always ≤ 1:

$$\text{demand factor} = \frac{\text{maximum demand}}{\text{connected load}}$$

These are explained further in 2(a).

2. Answer

Requirements of a base load plant:

- Low operating (fuel) cost per kWh, since it supplies most of the energy; a high capital cost is acceptable.
- High efficiency at a steady, nearly full load, and the ability to run continuously for long periods.
- High reliability and low maintenance, with long intervals between shutdowns.
- Large unit size. Examples: large coal-fired steam stations, nuclear stations, run-of-river hydro.

Requirements of a peak load plant:

- Quick starting and fast loading (minutes), and the ability to follow rapid changes of load.
- Low capital cost per kW, since it runs only a few hours a year; a higher fuel cost is acceptable.
- Good efficiency at part load, and the ability to stand frequent starts and stops.
- Examples: gas turbines, diesel stations, pumped-storage and reservoir hydro stations.

Given:

- Plant rating (taken as the connected load) = 30 MW
- Maximum demand (annual peak load), P_{max} = 25 MW
- Individual maximum demands = 10, 8.5, 5 and 4.5 MW
- Annual load factor, LF = 45 % = 0.45
- Hours in a year = $365 \times 24 = 8760$ h

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Step 1: Average load

$$P_{av} = \text{LF} \times P_{max} = 0.45 \times 25 = 11.25 \text{ MW}$$

$$\boxed{P_{av} = 11.25 \text{ MW}}$$

On average over the year, the station delivers 11.25 MW.

Step 2: Energy supplied every year

$$E = P_{av} \times 8760 = 11.25 \times 8760 = 98\,550 \text{ MWh}$$

$$\boxed{E = 98\,550 \text{ MWh} = 9.855 \times 10^7 \text{ kWh}}$$

Step 3: Diversity factor

$$\text{Sum of maximum demands} = 10 + 8.5 + 5 + 4.5 = 28 \text{ MW}$$

$$\text{DF} = \frac{28}{25} = 1.12$$

$$\boxed{\text{DF} = 1.12}$$

The consumers do not all peak together, so the station needs only 25 MW, not 28 MW.

Step 4: Demand factor

$$\text{demand factor} = \frac{\text{maximum demand}}{\text{connected load}} = \frac{25}{30} = 0.833$$

$$\boxed{\text{demand factor} = 0.833}$$

Assumption: the paper gives no separate connected load, so the 30 MW rating of the station is taken as the connected load, as in V.K. Mehta.

Final Answer:

- Average load = 11.25 MW
- Energy supplied per year = 98 550 MWh (9.855×10^7 kWh)
- Diversity factor = 1.12
- Demand factor = 0.833

3. Notes and Remarks

- The **capacity factor** of this station is $11.25/30 = 0.375$, and its **reserve capacity** is $30 - 25 = 5$ MW.
- If the question meant the demand factor of the consumers' own installations, their connected load would be needed; it is not given.

4. Final Concise Answer

(a) Generating capacity rises with higher steam pressure and temperature, reheating, a lower condenser pressure, regenerative feed heating, an economiser and air preheater, good combustion, larger units, a high load factor with good maintenance, and combined cycle working.

(b) A fuel is judged by its calorific value, moisture, volatile matter, fixed carbon, ash, sulphur, ignition temperature, ease of handling and storage, and its cost and availability.

(c) In a combined cycle, the hot exhaust of a gas turbine raises steam in a heat-recovery boiler for a steam turbine, so one fuel drives two generators at about 55–60 % efficiency.

(d) A base load plant needs a low running cost and steady, reliable, efficient operation; a peak load plant needs quick starting and a low capital cost. Average load = 11.25 MW, energy = 98 550 MWh per year, diversity factor = 1.12, demand factor = 0.833.

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2. (a) What is the importance of load factor and diversity factor? Marks: 3

(b) Why is the load on a power station variable? What are the effects of variable load on the operation of the power station? Marks: 4

(c) Show that for the economic load sharing, the incremental heat rates are equal. Marks: 10

(d) The incremental fuel costs for two generating units A and B of a power plant are given by the following equations: $\frac{dF_A}{dP_A} = 0.06P_A + 11.4$, $\frac{dF_B}{dP_B} = 0.07P_B + 10$, where P is in mega-watts and F is in takas per hour.

(i) Find the economic loading of the two units when the total load to be supplied by the power station is 150 MW.

(ii) Find the loss in fuel cost per hour if the load is equally shared by the two units. Marks: 8

2.(a) What is the importance of load factor and diversity factor? Marks: 3

1. Theory

1. Fixed cost and running cost (Core Idea)

Every power station has two kinds of cost:

- **Fixed cost:** interest on the money spent to build it, depreciation, salaries. It is paid every year whether the station runs or not, and it depends on the **size** (MW) of the plant, which must be big enough for the **maximum demand**.
- **Running cost:** mainly fuel. It depends on the **energy** (kWh) generated.

The cost of one kWh is the fixed cost spread over all the kWh sold, plus the running cost of that kWh. So the more kWh a plant of a given size sells, the cheaper each kWh becomes. Load factor and diversity factor both measure how well the plant's size is used.

2. The two factors on a load curve

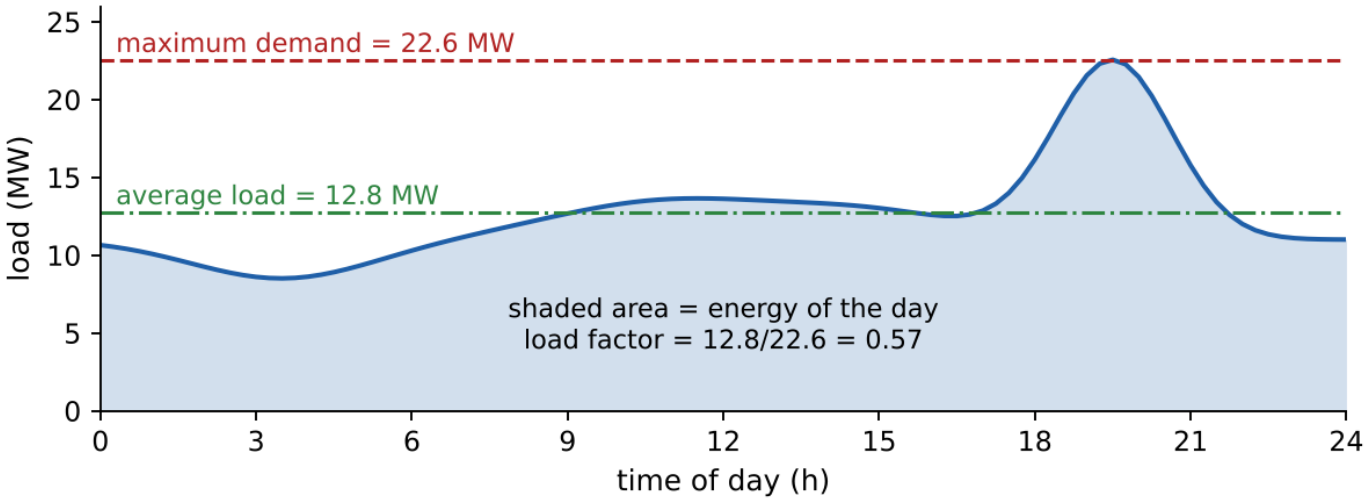


Figure 4: Load factor: the average load compared with the maximum demand

Reading the figure. The shaded area under the curve is the energy of the day. If the same energy were spread evenly, the load would be the dash-dot green line (average load). The dashed red line is the maximum demand, which sets the plant size. The closer the dash-dot line is to the dashed one, the higher the load factor.

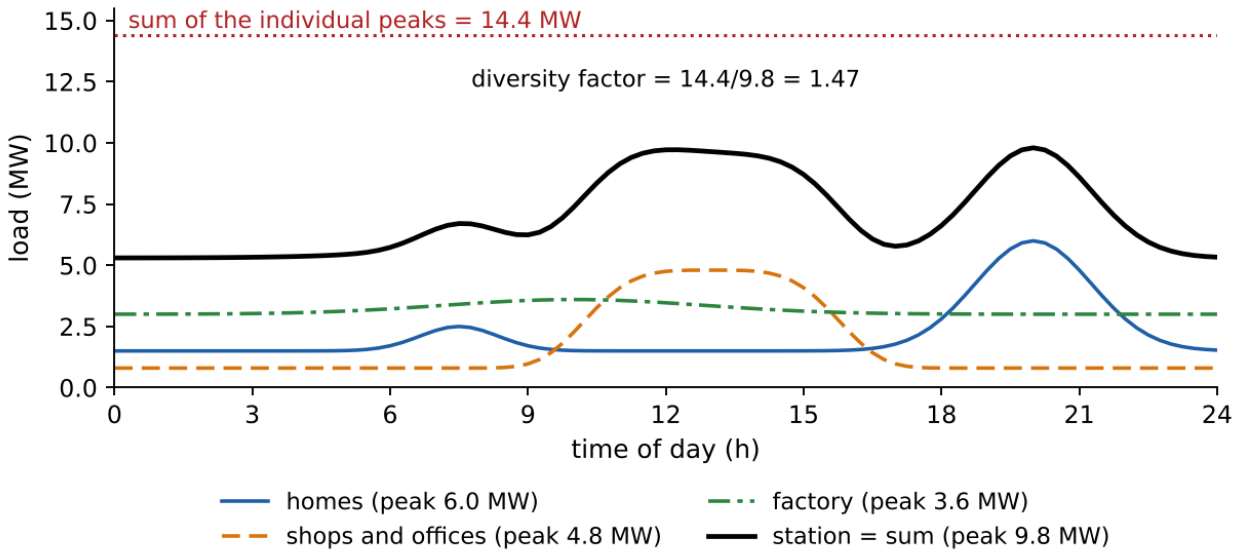


Figure 5: Diversity: three consumers that peak at different times

Reading the figure. Homes peak in the evening, shops at midday and the factory is steady. The station's load (thick solid line) is their sum, and its peak (9.8 MW) is much lower than the sum of their separate peaks (14.4 MW, red dotted line). The ratio is the diversity factor.

2. Answer

Importance of load factor (the average load divided by the maximum demand):

- The plant must be built for the maximum demand, but earns money only on the energy it sells. A **high load factor** means the plant's capacity is used for more hours, so the fixed cost is shared by more kWh and the **cost per kWh falls**.
- A high load factor lets the plant run near its rated output, where its efficiency is best.
- Supply companies encourage a high load factor with tariffs (for example a lower rate for night use).

Importance of diversity factor (the sum of the individual maximum demands divided by the maximum demand on the station):

- Because consumers do not all draw their maximum at the same time, the station needs a **smaller capacity** than the sum of their demands. A **high diversity factor** therefore reduces the plant size and its fixed cost, and so the **cost per kWh**.
- Mixing different types of load (homes, offices, industry) raises the diversity factor.

2.(b) Why is the load on a power station variable? What are the effects of variable load on the operation of the power station? Marks: 4

1. Theory

1. What the station must do (Core Idea)

Electricity cannot be stored in large amounts, so at every moment the station must generate exactly what the consumers are drawing. When consumers switch loads on or off, the station's output has to follow. This is why the load curve of 1(d) is the station's working schedule.

2. Answer

Why the load is variable: the load on a power station is the sum of the demands of all its consumers, and each consumer switches loads on and off at will. Domestic load peaks in the evening, commercial load in the day, irrigation and industrial loads follow their own seasons and shifts; the weather (heat,

cold, daylight) and holidays change all of them. So the total load varies from hour to hour, from day to day and from season to season.

Effects of variable load on the operation of the station:

1. **Need for additional equipment:** the station must have enough generating capacity for the peak, so some machines stand idle most of the time. Extra units (and spare boiler capacity) are installed only for the peak.
2. **Increased cost of production:** the idle capacity still costs interest and depreciation, and machines running at part load are less efficient, so the **cost per kWh rises**.
3. **Operating problems:** machines must be started, stopped and loaded up and down, and fuel and steam supply must be adjusted all the time, which needs careful scheduling and skilled staff.
4. **Effect on the plant itself:** frequent changes of load and temperature cause thermal stresses and wear; the voltage and frequency must be held steady by governors and regulators.

2.(c) Show that for the economic load sharing, the incremental heat rates are equal. Marks: 10

1. Theory

1. The input–output curve of a unit (Core Idea)

Each generating unit needs a certain **heat input** I (kcal/h, from its fuel) to give an electrical output P (MW). The graph of I against P is its **input–output curve**. It is not a straight line: it rises faster at high output, because losses grow with load. Multiplying the heat by the fuel price gives the **fuel cost** F (Tk/h).

2. Heat rate and incremental heat rate

- **Heat rate:** heat input per unit output, I/P (kcal/kWh). It says how much heat each kWh costs on average.
- **Incremental heat rate:** the slope of the input–output curve, dI/dP. It says how much **extra** heat is needed for *one more* MW of output from the present load. In money terms it is the **incremental fuel cost** dF/dP (Tk/MWh).

Because the curve gets steeper with output, the incremental rate rises with load: each extra MW costs more than the last.

3. The idea behind economic sharing

Suppose a station has two units and must supply a fixed total load. If unit 1 is costing more for its last MW than unit 2, take that MW off unit 1 and give it to unit 2: the total is unchanged, but the cost falls. Keep doing this until no shift saves anything. That happens when both units have the *same* incremental cost.

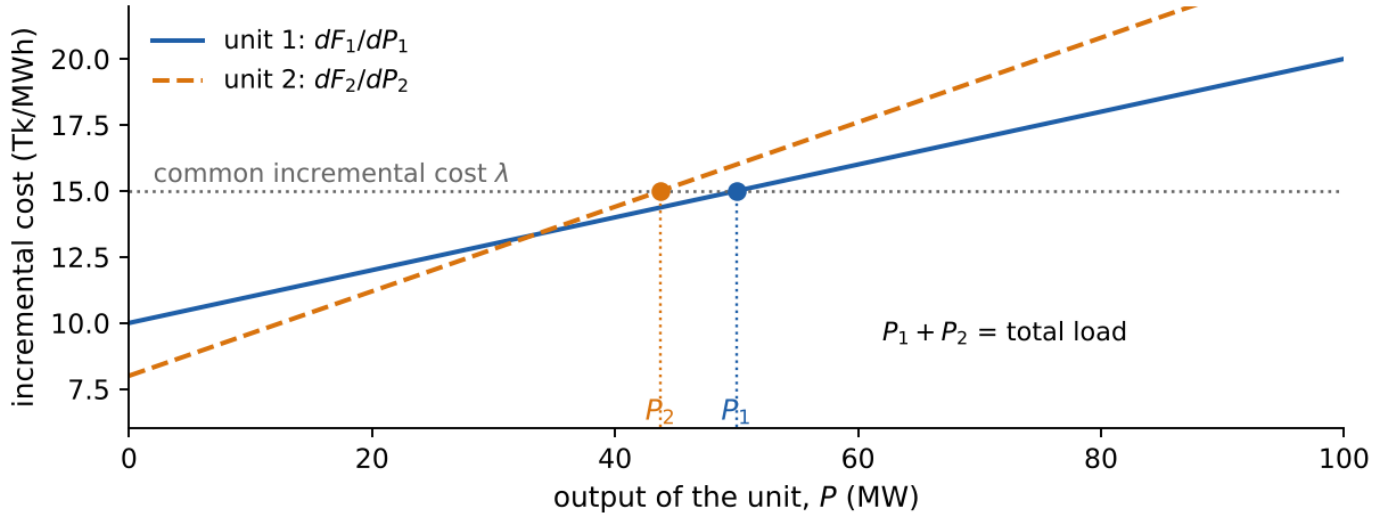


Figure 6: Economic sharing: both units work where their incremental costs are equal

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Reading the figure. The solid blue line is unit 1's incremental cost and the dashed orange line is unit 2's; both rise with output. A horizontal line at one common value λ cuts them at P_1 and P_2. The economic loading is the pair that has the same incremental cost and adds up to the total load.

2. Answer

To show: when a load is shared economically between units, their incremental heat rates (or incremental fuel costs) are equal.

Setup: let two units supply a total load P (the result extends to any number of units). Let

- P_1, P_2 = outputs of units 1 and 2 (MW), with $P_1 + P_2 = P$ = constant
- $I_1(P_1)$, $I_2(P_2)$ = heat inputs (kcal/h); for one fuel, the fuel costs are $F_1 = kI_1$ and $F_2 = kI_2$, where k is the cost of the fuel per kcal
- $I = I_1 + I_2$ = total heat input, which is to be a **minimum**

Step 1: Write the condition for a minimum

Since P is fixed, any increase in P_1 must be matched by an equal decrease in P_2:

$$P_1 + P_2 = P \quad \Rightarrow \quad dP_1 + dP_2 = 0 \quad \Rightarrow \quad dP_2 = -dP_1$$

The total input I is a minimum when a small change of the sharing does not change it:

$$dI = \frac{dI_1}{dP_1}\,dP_1 + \frac{dI_2}{dP_2}\,dP_2 = 0$$

Step 2: Substitute the constraint

$$\frac{dI_1}{dP_1}\,dP_1 - \frac{dI_2}{dP_2}\,dP_1 = 0 \quad \Rightarrow \quad \left(\frac{dI_1}{dP_1} - \frac{dI_2}{dP_2}\right) dP_1 = 0$$

Since $dP_1 \neq 0$,

$$\boxed{\frac{dI_1}{dP_1} = \frac{dI_2}{dP_2}}$$

That is, the incremental heat rates of the two units are equal. Multiplying both sides by k gives the same in money: $\dfrac{dF_1}{dP_1} = \dfrac{dF_2}{dP_2}$.

Step 3: Check that it is a minimum, not a maximum

$$\frac{d^2I}{dP_1^2} = \frac{d^2I_1}{dP_1^2} + \frac{d^2I_2}{dP_2^2} > 0$$

This holds because each input–output curve gets steeper with load (rising incremental rate), so the equal-rate point is a minimum.

Step 4: The same result by reasoning

If the rates were unequal, say $\dfrac{dI_1}{dP_1} > \dfrac{dI_2}{dP_2}$, moving a small load ΔP from unit 1 to unit 2 would save $\dfrac{dI_1}{dP_1}\Delta P$ on unit 1 and cost only $\dfrac{dI_2}{dP_2}\Delta P$ on unit 2. The total input would fall, so the first sharing was not the cheapest. Such shifts stop saving only when the two rates are equal.

Step 5: Any number of units

For n units with $\sum P_i = P$, the same steps (or a Lagrange multiplier λ) give

$$\frac{dI_1}{dP_1} = \frac{dI_2}{dP_2} = \cdots = \frac{dI_n}{dP_n} = \lambda$$

Result: for economic load sharing, every unit runs at the same incremental heat rate λ (and the outputs add up to the load). As the load rises, λ rises, and each unit takes its share along its own

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curve.

3. Notes and Remarks

- The rule assumes every unit is running between its minimum and maximum outputs. A unit that reaches its limit is held there, and the rest share the remaining load at equal rates.
- Equal **heat rates** (I/P) are *not* the condition: a unit can have a good average heat rate and still be the dearer one for the next MW.
- In a whole power system, transmission losses are added and the condition becomes "equal incremental cost of power delivered".

2.(d) The incremental fuel costs for two generating units A and B of a power plant are given by the following equations: $\frac{dF_A}{dP_A} = 0.06P_A + 11.4$, $\frac{dF_B}{dP_B} = 0.07P_B + 10$, where P is in mega-watts and F is in takas per hour. (i) Find the economic loading of the two units when the total load to be supplied by the power station is 150 MW. (ii) Find the loss in fuel cost per hour if the load is equally shared by the two units. Marks: 8

1. Answer

Given:

- $\frac{dF_A}{dP_A} = 0.06P_A + 11.4$ Tk/MWh, $\frac{dF_B}{dP_B} = 0.07P_B + 10$ Tk/MWh
- Total load $P_A + P_B = 150$ MW

Step 1: Economic loading (i)

From 2(c), the cheapest sharing has equal incremental costs:

$$0.06P_A + 11.4 = 0.07P_B + 10, \qquad P_B = 150 - P_A$$

$$0.06P_A + 11.4 = 0.07(150 - P_A) + 10 = 20.5 - 0.07P_A$$

$$0.13P_A = 9.1 \quad \Rightarrow \quad P_A = 70 \text{ MW}, \qquad P_B = 150 - 70 = 80 \text{ MW}$$

$$\boxed{P_A = 70 \text{ MW}, \quad P_B = 80 \text{ MW}}$$

Check: $0.06(70) + 11.4 = 15.6$ and $0.07(80) + 10 = 15.6$ Tk/MWh, equal as required.

Step 2: Loss with equal sharing (ii)

With equal sharing, each unit carries 75 MW. Unit A goes up from 70 to 75 MW, and unit B goes down from 80 to 75 MW. The change in each unit's fuel cost is the area under its incremental-cost line.

Increase in the cost of unit A:

$$\Delta F_A = \int_{70}^{75} (0.06P_A + 11.4)\, dP_A = \Big[0.03P_A^2 + 11.4P_A\Big]_{70}^{75}$$

$$= 0.03(75^2 - 70^2) + 11.4(75 - 70) = 21.75 + 57 = 78.75 \text{ Tk/h}$$

Decrease in the cost of unit B:

$$\Delta F_B = \int_{75}^{80} (0.07P_B + 10)\, dP_B = \Big[0.035P_B^2 + 10P_B\Big]_{75}^{80}$$

$$= 0.035(80^2 - 75^2) + 10(80 - 75) = 27.125 + 50 = 77.125 \text{ Tk/h}$$

Net loss:

$$\Delta F = 78.75 - 77.125 = 1.625 \text{ Tk/h}$$

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Extra fuel cost = Tk 1.625 per hour

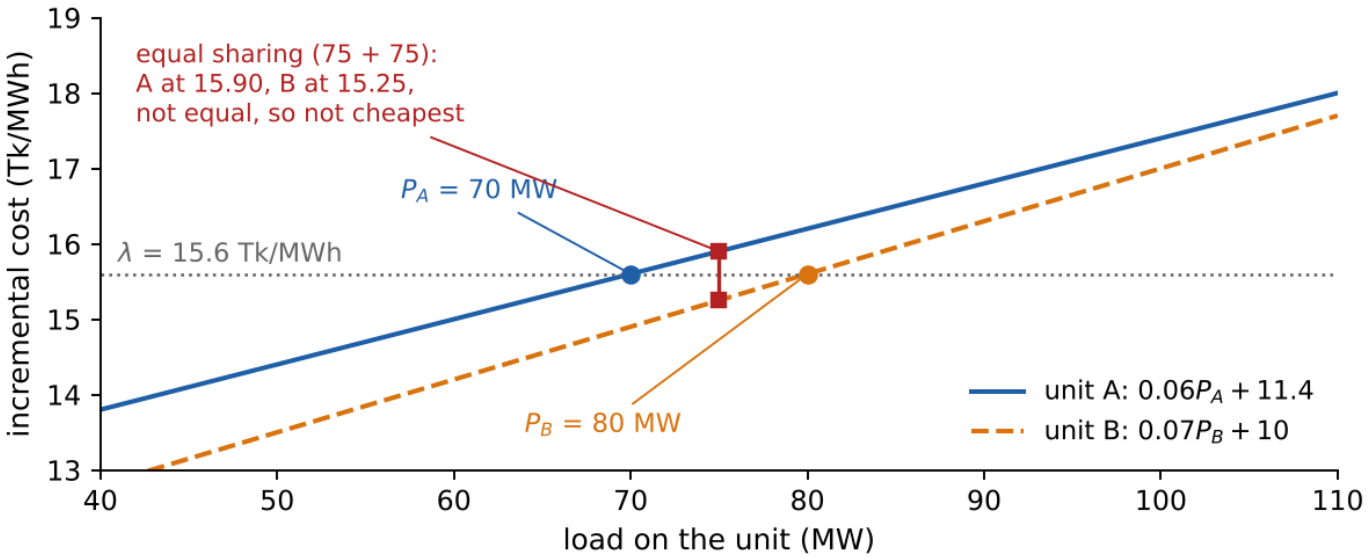


Figure 7: Incremental costs of units A and B: equal at 70 and 80 MW, unequal at 75 and 75 MW

Reading the figure. The two lines cross the common level 15.6 Tk/MWh at 70 MW (A, solid blue) and 80 MW (B, dashed orange): that is the economic loading. At an equal 75/75 sharing (the two squares), A works at 15.90 Tk/MWh but B at only 15.25, so moving load from A to B would still save money.

Final Answer:

- (i) Economic loading: $P_A = 70$ MW, $P_B = 80$ MW (incremental cost 15.6 Tk/MWh)
- (ii) Equal sharing costs Tk 1.625 more per hour

2. Notes and Remarks

The loss looks small, but it runs all year: $1.625 \times 8760 \approx$ Tk 14 235 per year for this one load level.

3. Final Concise Answer

(a) A high load factor spreads the fixed cost of a plant built for the maximum demand over more kWh, and a high diversity factor lets a smaller plant serve the same consumers; both lower the cost per kWh.

(b) The load varies because consumers switch their loads at will, by time of day, season and weather. It forces extra (often idle) capacity, raises the cost per kWh, complicates operation and stresses the plant.

(c) Minimising I_1+I_2 with P_1+P_2 fixed ($dP_2 = -dP_1$) gives $dI_1/dP_1 = dI_2/dP_2$; with rising incremental curves this is a minimum, and it extends to n units as $dI_i/dP_i = \lambda$.

(d) (i) $P_A = 70$ MW, $P_B = 80$ MW, at 15.6 Tk/MWh. (ii) Equal sharing loses Tk 1.625 per hour.

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3. (a) Draw the schematics of steam generating part of a coal fire power plant showing all basic components and direction of steam flow. Marks: 7

   (b) Differentiate between water tube boiler and fire tube boiler. Marks: 5

   (c) A superheater and an economizer improve the performance of a boiler-How? Marks: 5

   (d) The following data is supplied for a boiler plant: Boiler: Mass of coal supplied = 220 kg /hr. Calorific value of coal = 7200 kcal /kg. Mass of feed water = 200 kg/hr [*sic*: should be 2000 kg/hr]. Enthalpy of steam produced = 670 kcal/kg. Economizer: Inlet temperature of feed water = 18°C. Outlet temperature of water = 88°C. Atmospheric temperature = 19°C. Temperature of flue gas entering = 350°C. Mass of flue gases = 4150 kg/hr. Determine the following

   (i) Efficiency of boiler.
   (ii) Efficiency of economizer.
   (iii) Efficiency of whole boiler plant. Marks: 8

3.(a) Draw the schematics of steam generating part of a coal fire power plant showing all basic components and direction of steam flow. Marks: 7

1. Theory

1. What does the steam generating part do? (Core Idea)

The **steam generating part** (the boiler house) of a coal-fired station takes coal, air and water in, and sends out three things: superheated steam to the turbine, flue gas to the chimney, and ash to the ash pond. Inside it, four streams move side by side:

- **Coal:** from the store, crushed, ground to powder and blown into the furnace.
- **Air:** pushed in by a fan, warmed, and used to burn the coal.
- **Flue gas:** the hot gas from burning, which gives its heat to the water and air on its way to the chimney.
- **Water and steam:** water is heated, boiled and then superheated into dry, very hot steam.

2. The parts, in the order the water meets them

1. **Feed pump** pushes water into the boiler at high pressure.
2. **Economiser:** tubes in the cooler flue gas that warm the feed water.
3. **Steam drum:** a big horizontal drum where water and steam separate.
4. **Water walls:** tubes lining the furnace. Water runs down cooler pipes (downcomers) from the drum and rises up the hot wall tubes, boiling as it goes; the steam–water mixture returns to the drum.
5. **Superheater:** tubes in the hottest gas that heat the steam from the drum above its boiling point.

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2. Answer

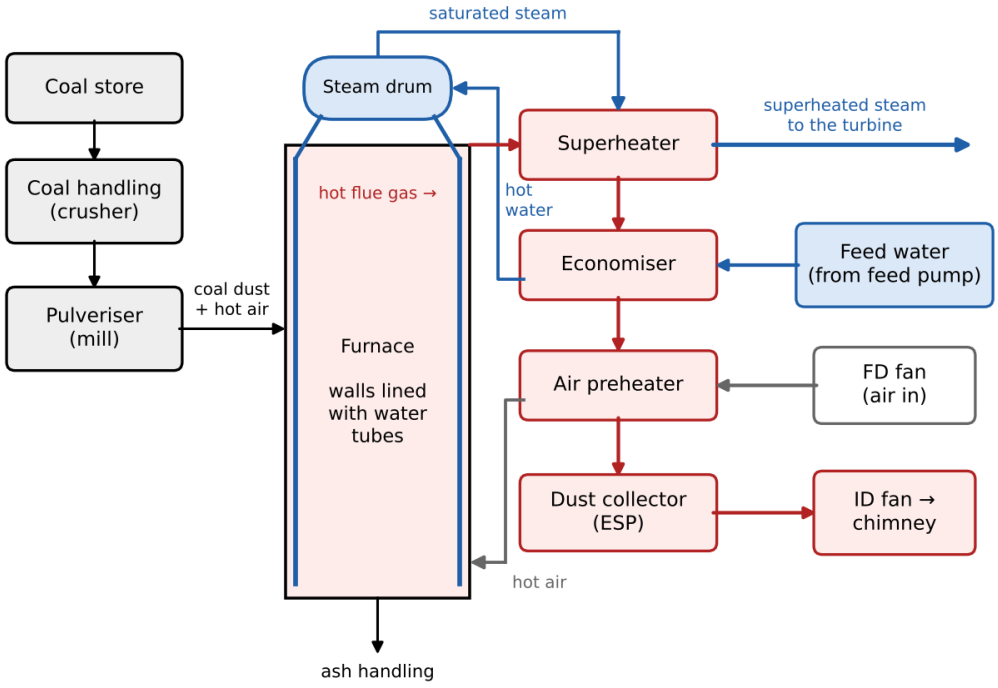


Figure 8: Steam generating part of a coal-fired power plant: coal, air, flue gas, and water and steam, each with its own arrows

Reading the figure. Coal goes down the left column and is blown, as dust, into the furnace. Feed water (right) enters the economiser, goes to the steam drum, boils in the furnace water walls and returns to the drum as steam; the saturated steam then passes through the superheater and leaves for the turbine. The flue gas leaves the furnace and passes the superheater, economiser, air preheater and dust collector before the ID fan sends it up the chimney. Air from the FD fan is heated in the air preheater and goes to the furnace.

The basic components:

- **Coal handling plant:** unloading, storing and crushing the coal.
- **Pulveriser (mill):** grinds the coal to a fine powder, which hot air carries to the burners (6(b)).
- **Furnace and water walls:** the coal burns in the furnace; the water-wall tubes absorb the heat and boil the water.
- **Steam drum:** separates steam from water; water returns to the walls.
- **Superheater:** raises the steam above its saturation temperature.
- **Economiser:** heats the feed water with the flue gas.
- **Air preheater:** heats the combustion air with the flue gas.
- **Forced-draught (FD) and induced-draught (ID) fans:** push the air in and pull the gas out.
- **Dust collector (electrostatic precipitator, ESP)** and **chimney:** clean and discharge the flue gas.
- **Ash handling plant:** removes the ash from the bottom of the furnace and from the dust collector.

Direction of steam flow: feed water → economiser → steam drum → water walls (boiling) → steam drum → superheater → main steam pipe → turbine.

3.(b) Differentiate between water tube boiler and fire tube boiler. Marks: 5

1. Theory

1. Two ways to put fire and water together (Core Idea)

A boiler has to get heat from hot gas into water. There are two ways to arrange the tubes:

- **Fire-tube boiler:** the hot gas flows **inside** the tubes, and the tubes sit in a big shell full of water.
- **Water-tube boiler:** the water flows **inside** the tubes, and the hot gas flows **around** them.

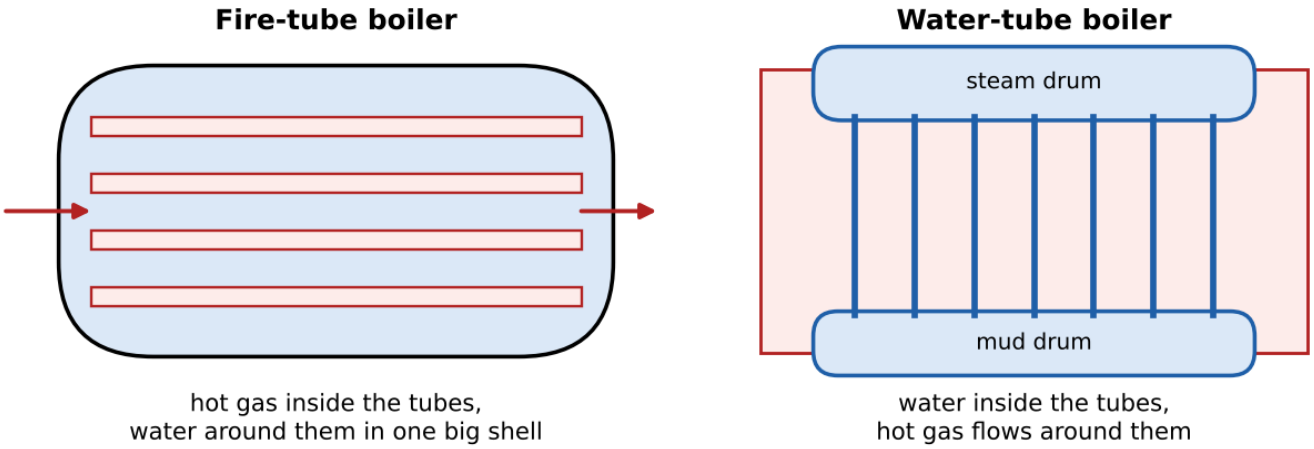


Figure 9: Fire-tube boiler (gas inside tubes) and water-tube boiler (water inside tubes)

Reading the figure. In the fire-tube boiler (left), the thin horizontal tubes carry hot gas through a shell full of water. In the water-tube boiler (right), the vertical water-filled tubes join a steam drum and a mud drum, and the hot gas fills the space around them.

2. Why the arrangement matters

The pressure a vessel can hold depends on its diameter: a thin tube holds high pressure easily, while a big shell must be very thick to do the same. In a water-tube boiler, the high-pressure water and steam are in thin tubes, so it can make high-pressure steam. In a fire-tube boiler, the whole shell is at steam pressure, which limits it to low pressures.

2. Answer

Point	Water-tube boiler	Fire-tube boiler
What is in the tubes	water and steam; hot gas outside	hot gas; water outside, in the shell
Steam pressure	high, up to about 250 bar	low, up to about 25 bar
Output	large: suits power stations	small: process steam, small plants
Water content	small, so it raises steam quickly	large, so it raises steam slowly
Response to load change	fast	slow, but steadier pressure
Danger on bursting	less: only a tube fails	more: a large mass of water flashes to steam
Size for given output	compact	bulky
Water quality	needs clean, treated water	less sensitive
Cost and upkeep	higher first cost; easy to replace tubes	cheaper and simpler

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Point	Water-tube boiler	Fire-tube boiler
Examples	Babcock and Wilcox, Stirling, modern drum boilers	Cochran, Lancashire, locomotive boilers

Table 2: Water-tube and fire-tube boilers compared

Conclusion: power stations use water-tube boilers because only they give the high pressure, large output and fast response a turbine needs.

3.(c) A superheater and an economizer improve the performance of a boiler-How? Marks: 5

1. Theory

1. Saturated and superheated steam (Core Idea)

When water boils at a given pressure, its temperature stays fixed (the **saturation temperature**, for example 100 °C at 1 atm, about 311 °C at 100 bar). Steam at that temperature is **saturated steam**; it usually carries tiny water droplets (wet steam). Heating the steam further, away from the water, raises its temperature above saturation: that is **superheated steam**. It is dry and holds more energy per kilogram.

2. The heat left in the flue gas

After passing the water walls and superheater, the flue gas is still at 350–450 °C. If it went straight up the chimney, that heat would be lost. The economiser and air preheater recover part of it.

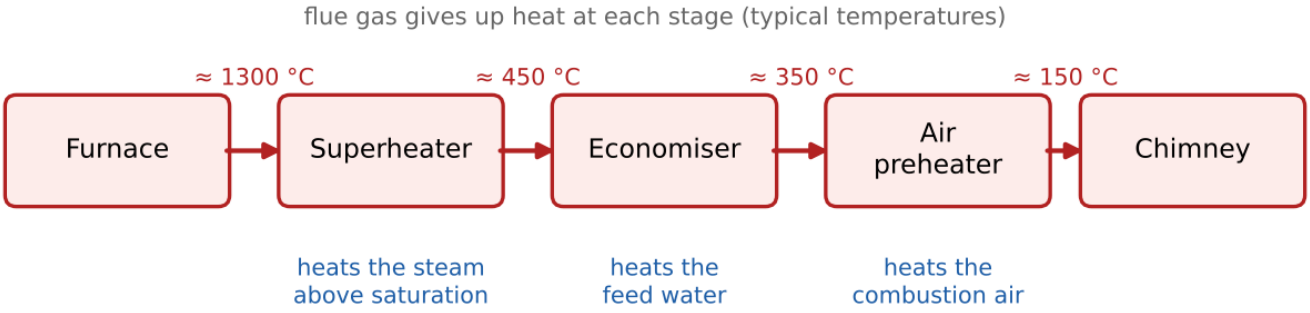


Figure 10: The flue gas cools as it passes each heat exchanger (typical temperatures)

Reading the figure. The flue gas moves from left to right and cools at each stage (temperatures above the arrows). Under each box is what that box heats with the gas: the superheater heats the steam, the economiser the feed water, and the air preheater the combustion air.

2. Answer

Superheater: a set of tubes placed in the hot gas path, through which the saturated steam from the drum passes and is heated above its saturation temperature. It improves performance because:

- the steam has more heat per kilogram, so the turbine gets **more work from each kg of steam** and the cycle efficiency rises;
- the steam is **dry**, so there are no water droplets to erode the turbine blades or cause losses in the last stages;
- less steam (and so a smaller boiler, pipes and condenser) is needed for the same output, and the steam does not condense in the pipes on its way to the turbine.

Economiser: a set of tubes placed in the flue gas after the boiler, through which the feed water passes

on its way to the drum. It improves performance because:

- it recovers heat from the flue gas that would otherwise go up the chimney, so the **boiler efficiency rises** (roughly 1 % for every 5–6 °C the feed water is heated) and **coal is saved**;
- the feed water enters the drum hotter, so the boiler raises steam faster and its output rises;
- hot feed water reduces the **thermal stress** on the drum and tubes that cold water would cause.

3.(d) The following data is supplied for a boiler plant: Boiler: Mass of coal supplied = 220 kg /hr. Calorific value of coal = 7200 kcal /kg. Mass of feed water = 200 kg/hr [*sic*: should be 2000 kg/hr]**. Enthalpy of steam produced = 670 kcal/kg. Economizer: Inlet temperature of feed water = 18°C. Outlet temperature of water = 88°C. Atmospheric temperature = 19°C. Temperature of flue gas entering = 350°C. Mass of flue gases = 4150 kg/hr. Determine the following (i) Efficiency of boiler. (ii) Efficiency of economizer. (iii) Efficiency of whole boiler plant. Marks: 8**

1. Theory

Note on the paper

The paper prints the mass of feed water as 200 kg/hr. With 220 kg/hr of coal, this would give a boiler efficiency of only 7.4 %, which is impossible for a real boiler. The value meant is **2000 kg/hr** (about 9 kg of steam per kg of coal, which is typical), and that is used below; the results for 200 kg/hr are in the Notes. The paper also leaves out two values that are needed. Notes written on the paper supply them: the enthalpy of the feed water entering the boiler, 88 kcal/kg, and the specific heat of the flue gas, 0.24 kcal/kg · °C.

1. Efficiency = useful heat out / heat in (Core Idea)

Every efficiency here has the same form: the heat that went where we wanted it, divided by the heat that was available.

- **Heat in from the coal:** mass of coal per hour × calorific value, $m_f\, C$ (kcal/h).
- **Heat taken up by water or steam:** mass × rise in enthalpy, $m_w(h_2 - h_1)$. For liquid water, the enthalpy measured from 0 °C is about $1 \times t$ kcal/kg (specific heat of water = 1 kcal/kg · °C), so water at 18 °C has $h = 18$ kcal/kg and at 88 °C, $h = 88$ kcal/kg.
- **Heat given up by the flue gas:** mass × specific heat × temperature drop, $m_g\, c_{pg}\, \Delta t$.

2. Three boundaries

- **Boiler alone:** water comes in from the economiser at 88 °C and leaves as steam at 670 kcal/kg; its heat source is the coal.
- **Economiser alone:** water is heated from 18 °C to 88 °C; its heat source is the flue gas. The heat available in the gas is counted down to the atmospheric temperature (19 °C), since that is the lowest temperature it could be cooled to.
- **Whole plant (boiler + economiser):** water enters at 18 °C and leaves as steam; the heat source is the coal.

2. Answer

Given:

- Mass of coal, $m_f = 220$ kg/h; calorific value, $C = 7200$ kcal/kg
- Mass of feed water (steam), $m_w = 2000$ kg/h (see the note above)
- Enthalpy of steam, $h = 670$ kcal/kg
- Feed water: enters the economiser at $t_1 = 18$ °C, leaves at $t_2 = 88$ °C, so it enters the boiler with $h_f = 88$ kcal/kg

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- Flue gas: $m_g = 4150$ kg/h, enters the economiser at $t_g = 350$ °C; atmospheric temperature $t_a = 19$ °C; $c_{pg} = 0.24$ kcal/kg · °C
- Specific heat of water, $c_w = 1$ kcal/kg · °C

Heat supplied by the coal:

$$Q_{in} = m_f\, C = 220 \times 7200 = 1\,584\,000 \text{ kcal/h}$$

Step 1: Efficiency of the boiler (i)

$$\eta_{boiler} = \frac{m_w\,(h - h_f)}{m_f\, C} = \frac{2000\,(670 - 88)}{1\,584\,000} = \frac{1\,164\,000}{1\,584\,000} = 0.7348$$

$$\boxed{\eta_{boiler} = 73.48\ \%}$$

Nearly three quarters of the coal's heat goes into turning the 88 °C water into steam.

Step 2: Efficiency of the economiser (ii)

Heat gained by the water:

$$Q_w = m_w\, c_w\,(t_2 - t_1) = 2000 \times 1 \times (88 - 18) = 140\,000 \text{ kcal/h}$$

Heat available in the flue gas (above atmospheric temperature):

$$Q_g = m_g\, c_{pg}\,(t_g - t_a) = 4150 \times 0.24 \times (350 - 19) = 329\,676 \text{ kcal/h}$$

$$\eta_{eco} = \frac{Q_w}{Q_g} = \frac{140\,000}{329\,676} = 0.4247$$

$$\boxed{\eta_{eco} = 42.47\ \%}$$

The economiser recovers about 42 % of the heat the gas still carries when it reaches it.

Step 3: Efficiency of the whole boiler plant (iii)

$$\eta_{plant} = \frac{m_w\,(h - c_w t_1)}{m_f\, C} = \frac{2000\,(670 - 18)}{1\,584\,000} = \frac{1\,304\,000}{1\,584\,000} = 0.8232$$

$$\boxed{\eta_{plant} = 82.32\ \%}$$

Check: $1\,164\,000 + 140\,000 = 1\,304\,000$ kcal/h, so the plant's useful heat is the boiler's plus the economiser's.

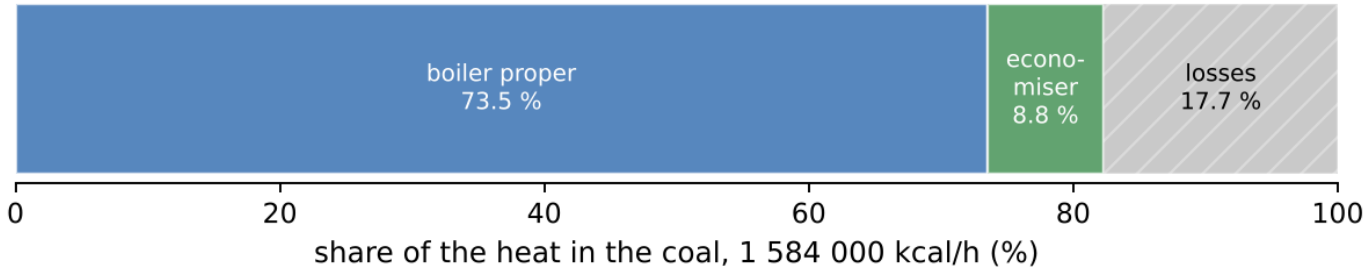


Figure 11: Where the heat of the coal goes

Reading the figure. Of the 1 584 000 kcal/h in the coal, 73.5 % is taken up in the boiler and 8.8 % in the economiser, which together make the 82.3 % plant efficiency; the remaining 17.7 % is lost (chimney, ash, radiation).

Final Answer:

- (i) Efficiency of the boiler = 73.48 %
- (ii) Efficiency of the economiser = 42.47 %

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- (iii) Efficiency of the whole boiler plant = 82.32 %

3. Notes and Remarks

- **With the printed 200 kg/h:** $\eta_{boiler} = 200(670-88)/1\,584\,000 = 7.35$ %, $\eta_{eco} = 14\,000/329\,676 = 4.25$ %, $\eta_{plant} = 200(670-18)/1\,584\,000 = 8.23$ %. These are far below any real boiler (70–90 %), which is why 2000 kg/h is taken as intended.
- The economiser lowers the gas temperature by $140\,000/(4150 \times 0.24) = 140.6$ °C, from 350 °C to about 209 °C.
- The economiser adds $82.32 - 73.48 = 8.84$ percentage points to the plant efficiency: this is the improvement described in 3(c).

4. Final Concise Answer

(a) Coal is stored, crushed and pulverised and blown into the furnace with hot air. Feed water passes through the economiser to the steam drum, boils in the furnace water walls, and the steam goes back to the drum, through the superheater and on to the turbine. The flue gas passes the superheater, economiser, air preheater and dust collector to the ID fan and chimney, and ash is removed below.

(b) A water-tube boiler has water in the tubes and gas outside: high pressure, large output, quick steaming, safer, for power stations. A fire-tube boiler has gas in the tubes and water in a shell: low pressure, small output, slow steaming, cheaper.

(c) The superheater dries the steam and raises its temperature, so each kg does more work and the turbine is protected from droplets. The economiser recovers flue-gas heat into the feed water, raising the boiler efficiency, saving coal and reducing thermal stress.

(d) With 2000 kg/h of feed water: boiler efficiency 73.48 %, economiser efficiency 42.47 %, whole plant 82.32 %.

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4. (a) Define plant use factor and plant utility factor. Marks: 4
   (b) Mention the types of reactors used in nuclear power plants. Marks: 5
   (c) A generating station has a maximum demand of 20 MW, a load factor of 60%, a plant capacity factor of 48% and a plant use factor of 80%. Find:
      (i) The daily energy produced
      (ii) The reserve capacity of the plant
      (iii) The maximum energy that could be produced daily if the plant was running all the time
      (iv) The maximum energy that could be produced daily if the plant was running fully loaded and operating as per schedule. Marks: 8
   (d) With diagram, write short notes on Kaplan and Francis turbine. Marks: 8

4.(a) Define plant use factor and plant utility factor. Marks: 4

1. Theory

1. Comparing what a plant did with what it could do (Core Idea)

A plant of capacity C (MW) could, in principle, produce C MW for every hour. Several factors compare its actual work with that limit; they differ in **which limit** they use:

- **Capacity factor:** actual energy compared with running at full capacity for *all* the hours of the period ($C \times 8760$ h for a year).
- **Plant use factor:** actual energy compared with running at full capacity for only the hours the plant was **actually in service**.
- **Utility (utilisation) factor:** the highest load compared with the capacity, that is, how much of the plant's size was ever needed.

The load factor, diversity factor and demand factor were built in 1(d).

2. Answer

Plant use factor: the ratio of the energy actually generated by the plant to the energy it could have generated running at full rated capacity during the hours it was in operation:

$$\text{plant use factor} = \frac{\text{station output in kWh}}{\text{plant capacity (kW)} \times \text{hours of use}}$$

For example, a 100 MW plant that ran 5000 hours in a year and generated 3×10^8 kWh has a plant use factor of $3 \times 10^8/(10^5 \times 5000) = 0.6$. It shows how heavily the plant is loaded while it runs; it is always $\geq$ the capacity factor.

Plant utility (utilisation) factor: the ratio of the maximum demand on the plant to its rated capacity:

$$\text{utility factor} = \frac{\text{maximum demand}}{\text{rated capacity of the plant}}$$

It shows how much of the installed capacity is actually used at the peak. A value near 1 means the plant is fully used (but has little reserve); a low value means part of the plant was built for nothing.

3. Notes and Remarks

V.K. Mehta calls the second one the **utilisation factor**; "utility factor" in the paper means the same.

4.(b) Mention the types of reactors used in nuclear power plants. Marks: 5

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1. Theory

1. How a reactor makes heat (Core Idea)

A **nuclear reactor** makes heat by **fission**: a slow neutron hits a uranium-235 nucleus, which splits into two smaller nuclei, releases a large amount of energy (about 200 MeV) and two or three new neutrons. These neutrons can split more nuclei, giving a **chain reaction**. The heat is carried away to raise steam, just as a furnace does in a coal plant.

2. The parts that make reactors differ

- **Fuel:** natural uranium (0.7 % U-235), enriched uranium (3–5 % U-235) or plutonium.
- **Moderator:** a material that slows the fast fission neutrons, because slow (thermal) neutrons cause fission in U-235 much more easily. Ordinary (light) water, heavy water (D_2O) and graphite are used.
- **Coolant:** the fluid that carries the heat out of the core: water, heavy water, CO_2 or helium gas, or liquid sodium.
- **Control rods:** rods of boron or cadmium that absorb neutrons; pushing them in slows the reaction.

Reactors are named mainly by their coolant and moderator.

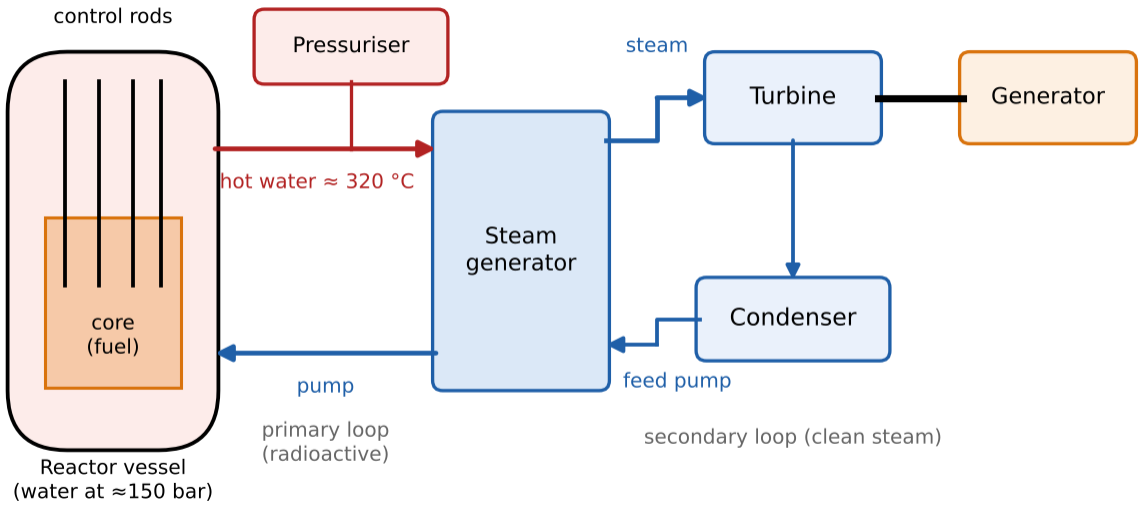


Figure 12: A pressurised water reactor: the primary loop carries heat to the steam generator, the secondary loop drives the turbine

Reading the figure. In the reactor vessel (left), the fuel core heats water that is kept liquid at about 150 bar by the pressuriser. This hot water passes through the steam generator and is pumped back: that is the radioactive primary loop. In the steam generator it boils separate water into clean steam, which drives the turbine and generator and is condensed and pumped back.

2. Answer

Nuclear reactors are classified first by the energy of the neutrons that cause fission:

- **Thermal reactors:** use a moderator to slow the neutrons; most power reactors.
- **Fast reactors:** no moderator; fast neutrons, used in breeder reactors.

The main types used in nuclear power plants:

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Reactor	Fuel	Moderator	Coolant
Pressurised water reactor (PWR)	enriched UO_2	light water	light water at high pressure (no boiling)
Boiling water reactor (BWR)	enriched UO_2	light water	light water, boiling in the core
Pressurised heavy water reactor (PHWR, CANDU)	natural uranium	heavy water	heavy water
Gas-cooled reactor (Magnox, AGR, HTGR)	natural or enriched uranium	graphite	CO_2 or helium
Fast breeder reactor (LMFBR)	plutonium + U-238	none	liquid sodium

Table 3: Types of reactors used in nuclear power plants

1. **PWR:** water under about 150 bar cools and moderates the core without boiling; a steam generator raises steam in a separate loop, so no radioactivity reaches the turbine. It is the most common type (Rooppur in Bangladesh uses a VVER-1200, a Russian PWR).
2. **BWR:** water boils in the core and the steam goes straight to the turbine; simpler, but the turbine becomes slightly radioactive.
3. **PHWR (CANDU):** heavy water is such a good moderator that natural uranium can be used, with no enrichment.
4. **Gas-cooled:** graphite moderator and CO_2 or helium coolant; gives higher steam temperatures.
5. **Fast breeder:** no moderator; it turns U-238 into new plutonium fuel and can produce more fuel than it burns.

4.(c) A generating station has a maximum demand of 20 MW, a load factor of 60%, a plant capacity factor of 48% and a plant use factor of 80%. Find: (i) The daily energy produced (ii) The reserve capacity of the plant (iii) The maximum energy that could be produced daily if the plant was running all the time (iv) The maximum energy that could be produced daily if the plant was running fully loaded and operating as per schedule. Marks: 8

1. Theory

This sub-part uses the factors defined in 1(d) and 4(a):

- Load factor $= \dfrac{\text{average load}}{\text{maximum demand}}$, so the average load is LF $\times$ maximum demand.
- Capacity factor $= \dfrac{\text{average load}}{\text{plant capacity}}$, so the plant capacity is average load / capacity factor.
- Plant use factor $= \dfrac{\text{energy}}{\text{capacity} \times \text{hours in operation}}$, which gives the hours the plant ran.
- **Reserve capacity** = plant capacity − maximum demand: the spare MW kept for breakdowns and growth.

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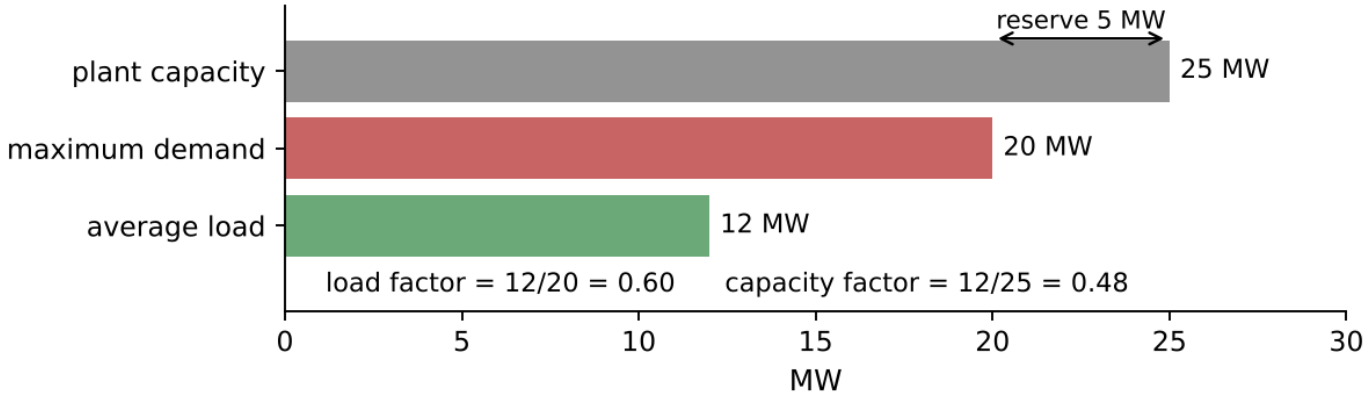


Figure 13: The three powers of the station and the factors that link them

Reading the figure. The plant is 25 MW (top bar), the highest load it met is 20 MW (middle bar), and the average load over the day is 12 MW (bottom bar). The 5 MW between capacity and maximum demand is the reserve; the ratios of the bars are the load factor and the capacity factor.

2. Answer

Given:

- Maximum demand, $P_{max} = 20$ MW
- Load factor, LF $= 0.60$; capacity factor, CF $= 0.48$; plant use factor, UF $= 0.80$
- One day $= 24$ h

Step 1: Daily energy produced (i)

$$P_{av} = \text{LF} \times P_{max} = 0.60 \times 20 = 12 \text{ MW}$$

$$E = P_{av} \times 24 = 12 \times 24 = 288 \text{ MWh}$$

$$\boxed{E = 288 \text{ MWh per day}}$$

Step 2: Reserve capacity (ii)

$$\text{plant capacity} = \frac{P_{av}}{\text{CF}} = \frac{12}{0.48} = 25 \text{ MW}$$

$$\text{reserve capacity} = 25 - 20 = 5 \text{ MW}$$

$$\boxed{\text{Reserve capacity} = 5 \text{ MW}}$$

Step 3: Maximum energy if the plant ran all the time (iii)

Running at its full 25 MW for all 24 hours:

$$E_{max} = 25 \times 24 = 600 \text{ MWh}$$

$$\boxed{E_{max} = 600 \text{ MWh per day}}$$

Step 4: Maximum energy if fully loaded as per schedule (iv)

The plant use factor gives the hours the plant was actually in operation:

$$\text{UF} = \frac{E}{\text{capacity} \times \text{hours in operation}} \quad \Rightarrow \quad \text{hours} = \frac{288}{25 \times 0.80} = 14.4 \text{ h}$$

Running fully loaded for those 14.4 hours:

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$$E_{sched} = 25 \times 14.4 = 360 \text{ MWh} \qquad \left(= \frac{E}{\text{UF}} = \frac{288}{0.8}\right)$$

$$\boxed{E_{sched} = 360 \text{ MWh per day}}$$

Final Answer:

- (i) Daily energy produced = 288 MWh
- (ii) Reserve capacity = 5 MW (plant capacity 25 MW)
- (iii) Maximum energy running all the time = 600 MWh per day
- (iv) Maximum energy running fully loaded as per schedule = 360 MWh per day

4.(d) With diagram, write short notes on Kaplan and Francis turbine. Marks: 8

1. Theory

1. How water turns a turbine (Core Idea)

A hydraulic turbine takes the energy of water falling through a height, the **head** H, and turns it into the rotation of a shaft. The power available is

$$P = \rho g Q H \quad \text{(watts)}$$

where

- ρ = density of water (1000 kg/m^3), g = 9.81 m/s^2
- Q = flow of water (m^3/s), H = head (m)

Turbines are of two kinds. In an **impulse** turbine (Pelton wheel), all the head is turned into a fast jet in air that strikes buckets. In a **reaction** turbine, the runner is full of water under pressure, and the water pushes on the blades both by its speed and by its pressure drop through the runner. Kaplan and Francis turbines are both reaction turbines.

2. The parts every reaction turbine has

- **Spiral (scroll) casing:** a pipe that wraps round the runner and gets narrower, so water enters evenly all the way round.
- **Guide vanes (wicket gates):** adjustable vanes that direct the water onto the runner at the right angle and control the flow (and so the power).
- **Runner:** the rotating wheel with blades, on the shaft that drives the generator.
- **Draft tube:** a widening tube from the runner outlet to the tail race, which regains part of the energy of the leaving water (5(c)).

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2. Answer

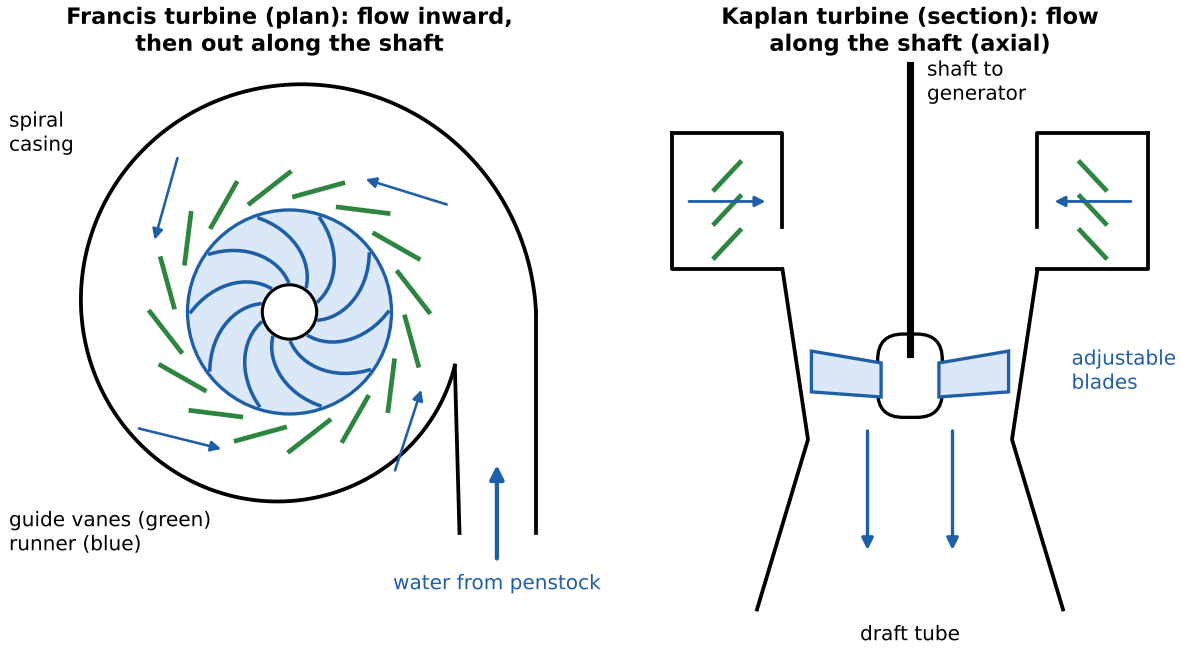


Figure 14: A Francis turbine seen from above, and a Kaplan turbine in section

Reading the figure. Left: in the Francis turbine, water from the penstock enters the spiral casing, passes through the ring of short straight guide vanes, and flows inward through the curved runner blades before turning down along the shaft. Right: in the Kaplan turbine, water passes the guide vanes, turns downward, and flows along the shaft through a propeller with a few adjustable blades, then leaves through the draft tube.

Francis turbine:

- An **inward mixed-flow reaction turbine**: water enters the runner radially from the outside, moves inward and leaves axially (along the shaft).
- **Parts:** spiral casing, adjustable guide vanes, runner with 16–24 fixed, curved blades, draft tube.
- **Working:** the guide vanes direct the water onto the runner blades; as the water moves inward through the curved passages, both its pressure and its velocity fall, and the reaction and impulse of the water on the blades turn the runner.
- **Use:** medium heads (about 30–300 m) and medium flows; the most widely used water turbine. Efficiency about 90 % or more.
- **Limitation:** fixed blades, so the efficiency falls noticeably at part load.

Kaplan turbine:

- An **axial-flow reaction turbine**: water flows parallel to the shaft through the runner.
- **Parts:** spiral casing, guide vanes, a runner like a ship's propeller with a large hub and only 3–8 blades, draft tube.
- **Special feature:** the runner blades are **adjustable**: a governor changes their angle together with the guide vanes, so the turbine stays efficient over a wide range of loads.
- **Use:** low heads (about 2–40 m) and large flows, such as river barrages; it runs at high speed, so a small generator can be used.
- **Efficiency:** about 90 %, and high even at part load.

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Point	Francis	Kaplan
Flow through runner	radial inward, then axial (mixed)	axial
Head	medium (30–300 m)	low (2–40 m)
Flow rate	medium	large
Runner blades	many (16–24), fixed	few (3–8), adjustable
Part-load efficiency	falls	stays high

Table 4: Francis and Kaplan turbines compared

3. Final Concise Answer

(a) Plant use factor = energy generated / (plant capacity × hours in operation). Plant utility (utilisation) factor = maximum demand / rated capacity of the plant.

(b) Thermal and fast reactors; in practice PWR, BWR, PHWR (CANDU), gas-cooled (Magnox, AGR, HTGR) and the fast breeder (LMFBR), told apart by fuel, moderator and coolant.

(c) Daily energy 288 MWh; plant capacity 25 MW, so the reserve is 5 MW; 600 MWh per day running all the time at full load; 360 MWh per day running fully loaded for the scheduled 14.4 h.

(d) The Francis turbine is a mixed-flow (radial in, axial out) reaction turbine with fixed blades for medium heads. The Kaplan turbine is an axial-flow propeller reaction turbine with adjustable blades for low heads and large flows, efficient over a wide load range.

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5. (a) Write short notes on the following type of tariffs:

   (i) Flat demand rate
   (ii) Straight line meter rate Marks: 5

   (b) Estimate the generating cost per KWh delivered from a generating station from the following data: Plant capacity = 50 MW. Annual load factor = 40%. Capital cost = 1.2 crores. Annual cost of wages, taxation, etc. = Taka 4 lakhs. Cost of fuel, lubrication, maintenance, etc. = 1.0 paisa/kWh generated. Interest 5% per annum, depreciation 6% per annum of initial value. Marks: 5

   (c) Write short note on-

   (i) Spillways
   (ii) Surge tanks
   (iii) Draft tubes Marks: 6

   (d) An electric supply company having a maximum load of 50 MW generates 18×10^7 units per annum and the supply consumers have an aggregate demand of 75 MW. The annual expenses including capital charges are: For fuel = Taka 90 lakhs. Fixed charges concerning generation = Taka 28 lakhs. Fixed charges concerning transmission and distribution = Taka 32 lakhs. Assuming 90% of the fuel cost is essential to running charges and the loss in transmission and distribution as 15% of KWh generated, deduce a two part tariff to find the actual cost of supply to the consumers. Marks: 9

5.(a) Write short notes on the following type of tariffs: (i) Flat demand rate (ii) Straight line meter rate Marks: 5

1. Theory

1. What a tariff is (Core Idea)

A **tariff** is the rule by which a supply company works out each consumer's electricity bill. A fair tariff must recover the company's three kinds of cost:

- **Fixed cost:** depends on the plant size, so on the consumer's **maximum demand** (kW).
- **Running cost:** depends on the **energy** used (kWh).
- **Customer cost:** meter reading, billing, the service line; the same for every consumer.

Different tariffs recover these in different ways. The two in this question are the simplest.

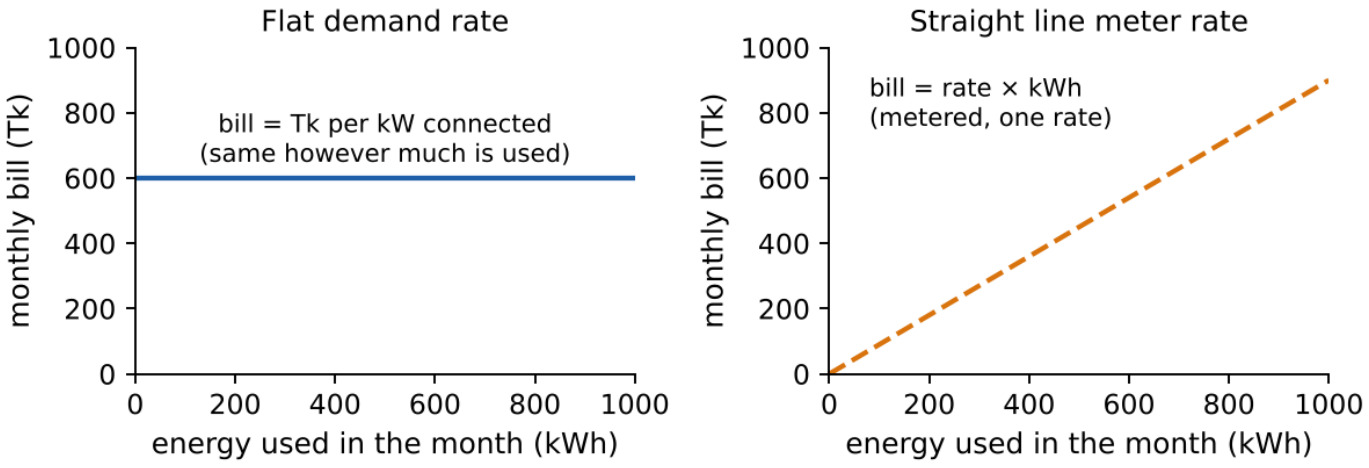


Figure 15: Monthly bill against energy used: flat demand rate and straight line meter rate

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Reading the figure. With the flat demand rate (left, solid blue), the bill is a flat line: it depends on the load connected, not on the kWh used. With the straight line meter rate (right, dashed orange), the bill rises in a straight line from zero with the kWh used.

2. Answer

(i) Flat demand rate: the bill depends only on the **maximum demand** (or the connected load) of the consumer, and not on the energy used. It is expressed as

$$\text{bill} = a \times \text{kW}$$

where a is a fixed charge in Tk per kW of connected load (or per lamp) per month.

- **Advantage:** no energy meter is needed, so metering and meter-reading costs are saved; the bill is easy to work out.
- **Disadvantage:** the consumer pays the same however much energy is used, so energy may be wasted; the supplier cannot tell the actual use.
- **Use:** street lighting, signal and sign lighting, and small loads whose hours of use are known.

(ii) Straight line meter rate: the bill is the **energy consumed** (kWh, read on a meter) times a fixed rate per kWh:

$$\text{bill} = b \times \text{kWh}$$

- **Advantage:** simple and easy to understand; each consumer pays in proportion to the energy used.
- **Disadvantages:** it does not charge for the maximum demand, so a consumer with a large demand but little use (low load factor) pays too little toward the fixed cost; a consumer who uses no energy pays nothing at all, though the company has spent money to serve them; and there is no reward for using more.

5.(b) Estimate the generating cost per KWh delivered from a generating station from the following data: Plant capacity = 50 MW. Annual load factor = 40%. Capital cost = 1.2 crores. Annual cost of wages, taxation, etc. = Taka 4 lakhs. Cost of fuel, lubrication, maintenance, etc. = 1.0 paisa/kWh generated. Interest 5% per annum, depreciation 6% per annum of initial value. Marks: 5

1. Theory

1. Annual cost and cost per kWh (Core Idea)

The cost of a kWh is found by adding up all the costs of a year and dividing by the kWh generated that year:

$$\text{cost per kWh} = \frac{\text{annual fixed cost}}{\text{annual energy (kWh)}} + \text{running cost per kWh}$$

- **Annual fixed cost:** interest on the capital (the money borrowed to build the plant is paid for every year), depreciation (money set aside to replace the plant when it is worn out, 6(a)), and wages and taxes.
- **Running cost:** fuel, lubrication and maintenance, which grow with the kWh.

2. Units used in Bangladesh and India

- 1 lakh = 10^5, 1 crore = 10^7; 1 taka = 100 paisa.
- Energy in a year at a load factor LF: E = capacity × LF × 8760 h (here the maximum demand is taken equal to the plant capacity).

2. Answer

Given:

- Plant capacity = 50 MW = 50 000 kW; annual load factor = 0.40
- Capital cost = 1.2 crore = Tk 1.2×10^7
- Interest = 5 % and depreciation = 6 % per annum of the capital cost
- Wages, taxation, etc. = Tk 4 lakh = Tk 4×10^5 per year
- Fuel, lubrication, maintenance, etc. = 1.0 paisa per kWh generated

Step 1: Energy generated per year

$$E = 50\,000 \times 0.40 \times 8760 = 1.752 \times 10^8 \text{ kWh}$$

Step 2: Annual fixed charges

$$\text{interest + depreciation} = (0.05 + 0.06) \times 1.2 \times 10^7 = \text{Tk } 13.2 \times 10^5$$

$$\text{total fixed charges} = 13.2 \times 10^5 + 4 \times 10^5 = \text{Tk } 17.2 \times 10^5 \text{ per year}$$

Step 3: Cost per kWh

$$\text{fixed cost per kWh} = \frac{17.2 \times 10^5}{1.752 \times 10^8} = \text{Tk } 0.00982 = 0.98 \text{ paisa}$$

$$\text{total} = 0.98 + 1.0 = 1.98 \text{ paisa per kWh}$$

$$\boxed{\text{Generating cost} \approx 1.98 \text{ paisa per kWh}}$$

Assumption: no losses are given between generation and delivery, so the kWh delivered are taken equal to the kWh generated.

Final Answer: the generating cost is about 1.98 paisa per kWh (0.98 paisa fixed + 1.0 paisa running).

5.(c) Write short note on- (i) Spillways (ii) Surge tanks (iii) Draft tubes Marks: 6

1. Theory

1. The layout of a hydro-electric plant (Core Idea)

A **dam** across a river stores water in a **reservoir**. The water is led through a tunnel and a steep pipe, the **penstock**, down to the **power house**, where it drives the turbines (4(d)), and then leaves by the **tail race** back to the river. The height between the reservoir level and the tail race is the head. The three parts in this question protect or improve this water path.

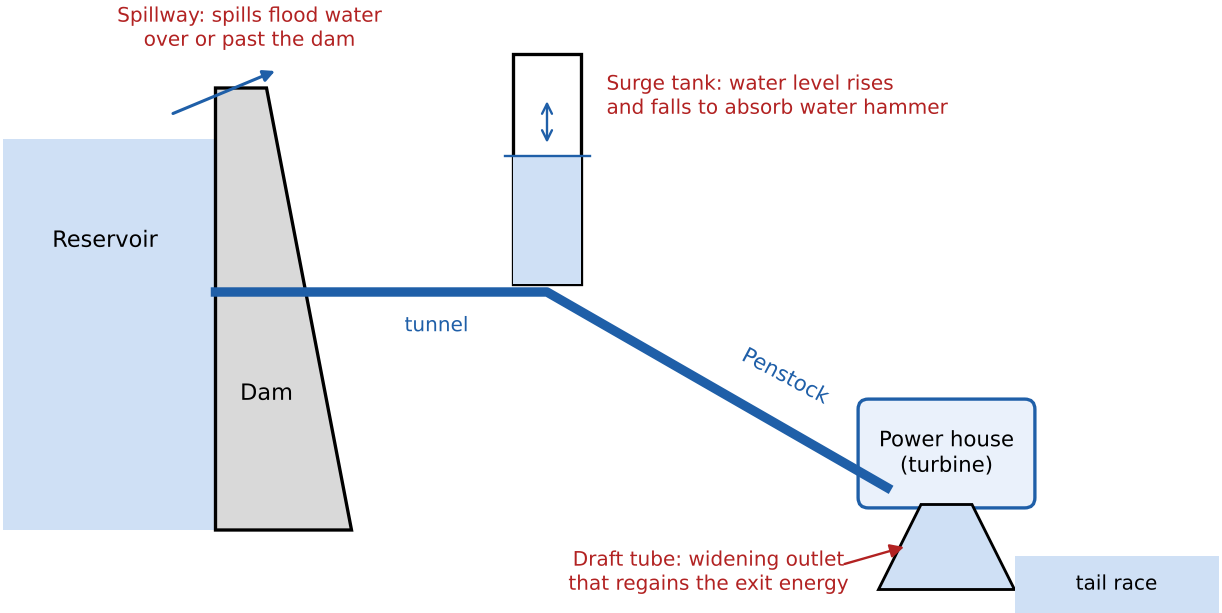


Figure 16: Layout of a hydro-electric plant with a spillway, a surge tank and a draft tube

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Reading the figure. Water flows from the reservoir (left) through the tunnel and penstock to the turbine in the power house, and out through the draft tube into the tail race. The spillway lets flood water pass over or past the dam; the surge tank stands on the pipe line just before the penstock; the draft tube is the widening outlet under the turbine.

2. Answer

(i) Spillway: a passage built on or beside a dam to discharge the surplus water of the reservoir safely downstream when floods raise the water level.

- **Purpose:** to stop the water from rising above the top of the dam (overtopping), which could destroy it, and to keep the reservoir at a safe level.
- **Types:** overflow (ogee) spillway over the dam crest, chute spillway, side-channel spillway, siphon spillway, shaft spillway; gates control the discharge.
- At its foot, energy dissipators (a stilling basin) stop the fast water from eroding the river bed.

(ii) Surge tank: an open vertical tank (or shaft) connected to the pipe line between the tunnel and the penstock, as close to the power house as possible.

- **Purpose:** when the load drops suddenly, the turbine gates close and the moving water in the long pipe is stopped; its momentum would cause a large pressure rise, called **water hammer**, which could burst the penstock. The surge tank lets the water rise in it instead, absorbing the surge.
- When the load rises suddenly, the tank supplies extra water at once, before the water in the long tunnel can speed up.
- It shortens the length of pipe that feels water hammer to the penstock alone. Types: simple, restricted-orifice and differential surge tanks.

(iii) Draft tube: an airtight pipe of gradually increasing area that connects the outlet of a reaction turbine to the tail race.

- **Purpose 1:** it lets the turbine be placed above the tail-race level (for easy access) without losing that part of the head, because the water column in the tube creates suction below the runner.
- **Purpose 2:** as the area widens, the water slows down, so much of the kinetic energy of the water leaving the runner is turned back into pressure; this increases the head available to the runner and the efficiency.
- **Types:** straight conical, elbow type (with a bend), Moody spreading tube, simple elbow tube.

5.(d) An electric supply company having a maximum load of 50 MW generates 18×10^7 units per annum and the supply consumers have an aggregate demand of 75 MW. The annual expenses including capital charges are: For fuel = Taka 90 lakhs. Fixed charges concerning generation = Taka 28 lakhs. Fixed charges concerning transmission and distribution = Taka 32 lakhs. Assuming 90% of the fuel cost is essential to running charges and the loss in transmission and distribution as 15% of KWh generated, deduce a two part tariff to find the actual cost of supply to the consumers. Marks: 9

1. Theory

Note on the paper

The mark of this sub-part is not printed clearly on the scan (only a handwritten 9 is visible). Nine marks make Question 5 total 25, like every other question, so 9 is used. The exponent in "18×10^7 units" is partly covered by handwriting; 18×10^7 is the value that fits the data (an annual load factor of 41 % on 50 MW).

1. Why two parts? (Core Idea)

The supply company's costs are of two kinds (5(a)):

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- **Fixed charges** depend on the size of plant and network needed, that is, on the consumers' **demand** in kW.
- **Running charges** depend on the **energy** supplied, in kWh.

A **two-part tariff** charges each kind to the quantity that causes it:

$$\text{bill} = A \times \text{kW of maximum demand} + B \times \text{kWh used}$$

where

- A = annual fixed charges / total maximum demand of the consumers (Tk per kW per year)
- B = annual running charges / kWh **sold** to the consumers (Tk per kWh)

2. Two points to watch

- The fixed charges are divided by the consumers' **aggregate demand** (75 MW), which is larger than the station's maximum load (50 MW) because of diversity (2(a)): every consumer pays for their own demand.
- The running charges are divided by the kWh that **reach the consumers**, after the transmission and distribution loss, because the lost kWh still had to be generated and paid for.

2. Answer

Given:

- Units generated = 18×10^7 kWh per year; loss in T&D = 15 % of kWh generated
- Aggregate maximum demand of the consumers = 75 MW = 75 000 kW
- Fuel = Tk 90 lakh, of which 90 % is running and 10 % fixed
- Fixed charges: generation Tk 28 lakh; transmission and distribution Tk 32 lakh

Step 1: Split the annual cost

$$\text{running charges} = 0.90 \times 90 = \text{Tk } 81 \text{ lakh}$$

$$\text{fixed charges} = 0.10 \times 90 + 28 + 32 = 9 + 28 + 32 = \text{Tk } 69 \text{ lakh}$$

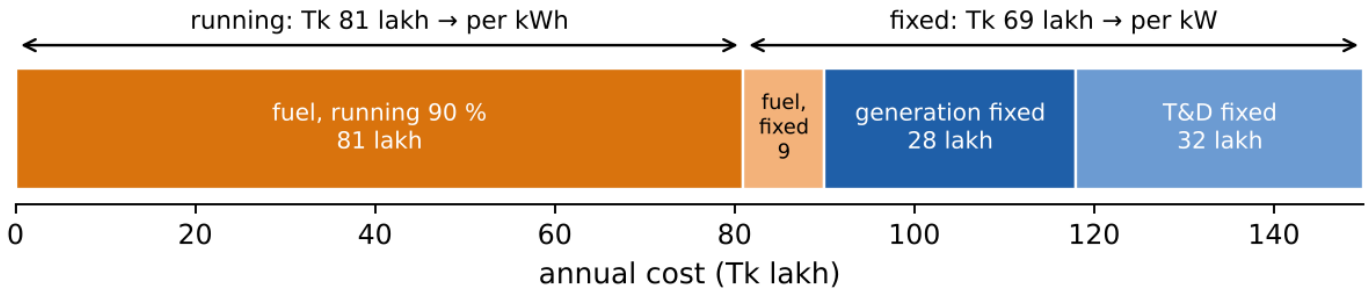


Figure 17: The annual cost split into running and fixed charges

Reading the figure. Of the Tk 150 lakh total, the running part (Tk 81 lakh, the long first bar) is recovered per kWh, and the fixed part (Tk 69 lakh: the small fixed share of the fuel cost and the generation and T&D bars) per kW of demand.

Step 2: Energy delivered to the consumers

$$E_{sold} = (1 - 0.15) \times 18 \times 10^7 = 15.3 \times 10^7 \text{ kWh}$$

Step 3: Charge per kW of maximum demand

$$A = \frac{69 \times 10^5}{75\,000} = \text{Tk } 92 \text{ per kW per year}$$

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Step 4: Charge per kWh

$$B = \frac{81 \times 10^5}{15.3 \times 10^7} = \text{Tk } 0.0529 = 5.29 \text{ paisa per kWh}$$

Step 5: The tariff

Tk 92 per kW of maximum demand per year + 5.29 paisa per kWh

A consumer with a maximum demand of D kW who uses E kWh in a year pays $92D + 0.0529E$ taka.
Final Answer: two-part tariff = Tk 92 per kW of maximum demand per annum + 5.29 paisa per kWh consumed.

3. Notes and Remarks

- Check: the tariff recovers $92 \times 75\,000 + 0.0529 \times 15.3 \times 10^7 = 69 + 81 = 150$ lakh taka, the whole annual cost.
- The average cost of supply works out at $150 \times 10^5 / 15.3 \times 10^7 = 9.8$ paisa per kWh sold.

4. Final Concise Answer

(a) Flat demand rate: bill = Tk per kW of connected load, independent of energy; no meter needed, used for street lighting, but invites waste. Straight line meter rate: bill = Tk per kWh × kWh used; simple and proportional to use, but ignores the maximum demand and the fixed cost.

(b) Energy = 1.752×10^8 kWh; fixed charges Tk 17.2 lakh, or 0.98 paisa/kWh; with 1.0 paisa running cost the generating cost is 1.98 paisa/kWh.

(c) A spillway safely passes surplus flood water past the dam. A surge tank on the pipe line absorbs water hammer when the load drops and supplies water when it rises. A draft tube, a widening outlet below a reaction turbine, regains the exit energy and lets the turbine sit above the tail race.

(d) Running charges Tk 81 lakh and fixed charges Tk 69 lakh; kWh sold 15.3×10^7; tariff = Tk 92 per kW per year + 5.29 paisa per kWh.

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6. (a) What are the methods of determining depreciation cost in power plants? Briefly explain one of the methods. Marks: 7
   (b) Why is pulverization done in a coal fired power plant? Give description of any basic type of pulverizers. Marks: 6
   (c) Write short notes on:
      (i) Impulse turbine
      (ii) Reaction turbines Marks: 6
   (d) Discuss the advantages and disadvantages of a Hydro-electric power plant. Marks: 6

6.(a) What are the methods of determining depreciation cost in power plants? Briefly explain one of the methods. Marks: 7

1. Theory

1. What is depreciation? (Core Idea)

A power plant wears out and grows out of date. After its **useful life** (20–40 years for most plant) it must be replaced, and by then it is worth only its scrap or **salvage value**. This steady loss of value is **depreciation**. A company that ignored it would one day have no money to buy the new plant. So, every year, part of the plant's cost is counted as a cost of running the station (it appears among the fixed charges in 5(b)) and set aside.

2. The quantities used

- P = initial (capital) cost of the plant
- S = salvage (scrap) value at the end of its life
- n = useful life in years
- i = annual rate of interest earned by money set aside

The total to be recovered over the life is $P - S$. The methods differ only in **how it is spread over the years**.

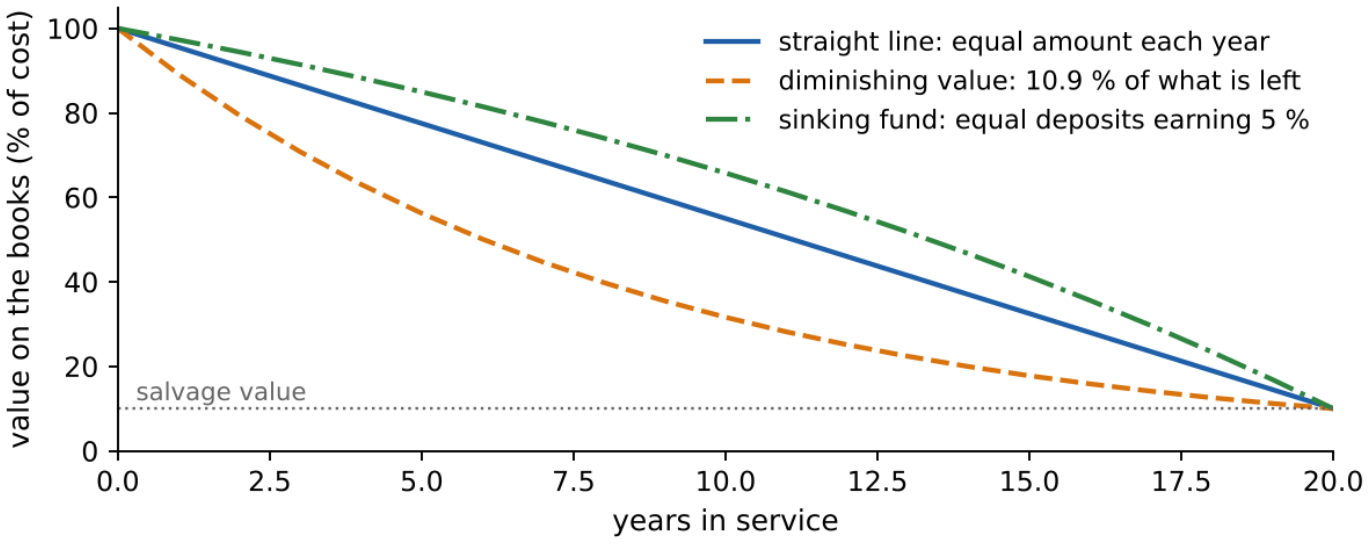


Figure 18: Value of a plant on the books over a 20-year life by the three methods (cost 100, salvage 10)

Reading the figure. All three curves start at the cost (100 %) and end at the salvage value (10 %) after 20 years. The straight-line value (solid blue) falls by the same amount each year; the diminishing-value curve (dashed orange) falls fast at first and slowly later; with the sinking fund (dash-dot green)

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the deposits are small at first and the interest they earn makes up the rest, so the book value falls slowly at first.

2. Answer

Depreciation is the decrease in the value of a power plant due to wear and tear, ageing and obsolescence. The methods of determining the annual depreciation cost are:

1. **Straight line method:** an equal amount is set aside every year: $D = \dfrac{P - S}{n}$.
2. **Diminishing value (reducing balance) method:** a fixed percentage r of the value left at the start of each year is set aside, so the charge is large in the early years and small later. The rate is chosen so that the value reaches S after n years: $P(1 - r)^n = S$.
3. **Sinking fund method:** a fixed sum is set aside every year and invested at compound interest, so that the deposits and their interest add up to $P - S$ at the end of the life.

Explanation of the sinking fund method:

Let q be the amount deposited at the end of each year at interest rate i. The first deposit earns interest for $n - 1$ years, the second for $n - 2$ years, and the last for none. The fund after n years is

$$q(1+i)^{n-1} + q(1+i)^{n-2} + \cdots + q(1+i) + q = q\,\frac{(1+i)^n - 1}{i}$$

This must equal the amount to be replaced, $P - S$:

$$q\,\frac{(1+i)^n - 1}{i} = P - S \quad \Rightarrow \quad \boxed{q = (P - S)\,\frac{i}{(1+i)^n - 1}}$$

where

- q = annual sinking-fund deposit (the annual depreciation charge)
- P, S, n, i as defined above

Example

for a plant costing Tk 100 lakh with a salvage value of Tk 10 lakh after 20 years, and money earning 5 %:

$$q = 90 \times \frac{0.05}{(1.05)^{20} - 1} = 90 \times \frac{0.05}{1.6533} = \text{Tk } 2.72 \text{ lakh per year}$$

against Tk 4.5 lakh per year by the straight-line method, because the interest earned does part of the saving.

Merits: it gives a uniform annual charge and takes account of the interest earned on the fund, so it is the most rational method and the one generally used for power plants. **Demerit:** the calculation needs an assumed interest rate.

6.(b) Why is pulverization done in a coal fired power plant? Give description of any basic type of pulverizers. Marks: 6

1. Theory

1. Why size matters in burning (Core Idea)

Coal burns only at its **surface**, where oxygen can reach it. Grinding a lump into dust multiplies the surface many times: a 1 cm cube cut into 75 µm grains has over 100 times more surface. Coal dust carried by air therefore burns almost like a gas: quickly, completely and with a flame that is easy to control.

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2. Answer

Pulverisation is grinding coal into a very fine powder (about 70–80 % passing a 75 µm, that is 200-mesh, sieve) and blowing it into the furnace with air, where it burns in suspension. It is done because:

1. **More complete combustion:** the huge surface area lets the coal burn completely with little excess air, so the **boiler efficiency rises** and less carbon is lost in the ash.
2. **Any grade of coal can be burnt,** including low-grade and high-ash coals that are hard to burn on a grate.
3. **Easy, quick control:** the firing rate can be changed like a gas or oil burner, so the boiler follows load changes quickly and starts fast.
4. **Large boilers:** pulverised firing is the only practical way to fire the very large boilers of modern stations; there is no grate and no clinker trouble.
5. **Higher temperature and less labour:** preheated air can be used, ash handling is easier, and the system is fully mechanised.

A basic type of pulveriser: the ball mill

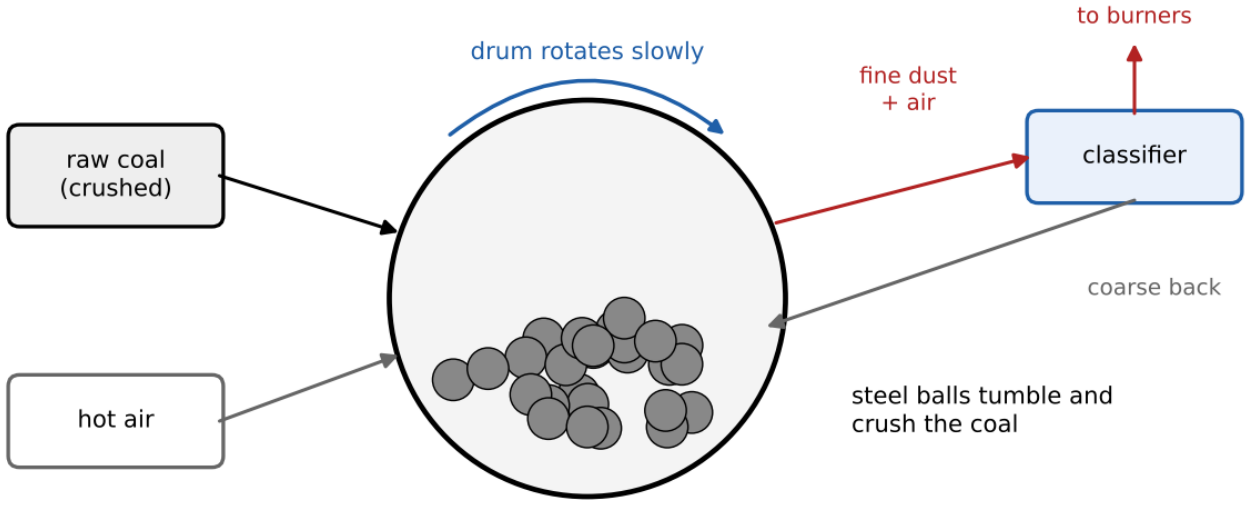


Figure 19: A ball mill: a rotating drum in which steel balls grind the coal

Reading the figure. Crushed coal and hot air enter the drum (left). As the drum turns, the steel balls tumble and crush the coal. The air carries the fine dust to the classifier; fine particles go on to the burners, and coarse ones return to the drum.

- **Construction:** a horizontal steel drum, lined with hard steel plates, about one-third filled with steel balls of different sizes (25–50 mm). It rotates slowly, at about 20 rpm.
- **Working:** crushed coal is fed in. As the drum rotates, the balls are carried up the side and fall back, crushing and rubbing the coal between them. Hot air from the air preheater dries the coal and carries the fine dust out of the drum to a **classifier**, which sends fine particles to the burners and returns oversize particles for regrinding.
- **Advantages:** simple and reliable, works with any coal including hard and abrasive coal, and keeps a store of coal in the drum.
- **Disadvantages:** heavy and bulky, uses more power per tonne than other mills, and is noisy.

Other types are the **ball-and-race mill** and the **bowl (roller) mill**, which grind the coal between rollers or balls and a rotating ring.

6.(c) Write short notes on: (i) Impulse turbine (ii) Reaction turbines Marks: 6

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1. Theory

1. Two ways to push a blade (Core Idea)

A steam turbine turns the energy of high-pressure steam into rotation. The steam must first be made to move fast, because only moving steam can push on a blade. There are two ways:

- **Impulse:** the steam expands in **fixed nozzles**, where all its pressure drop turns into a fast jet. The jet then hits the moving blades and is turned round; the change of its momentum pushes the blades, like water from a hose on a plate.
- **Reaction:** the **moving blades themselves are shaped like nozzles**. The steam expands and speeds up as it passes between them, and the blades are pushed back by the reaction of the jet leaving them, like a rocket or a garden sprinkler.

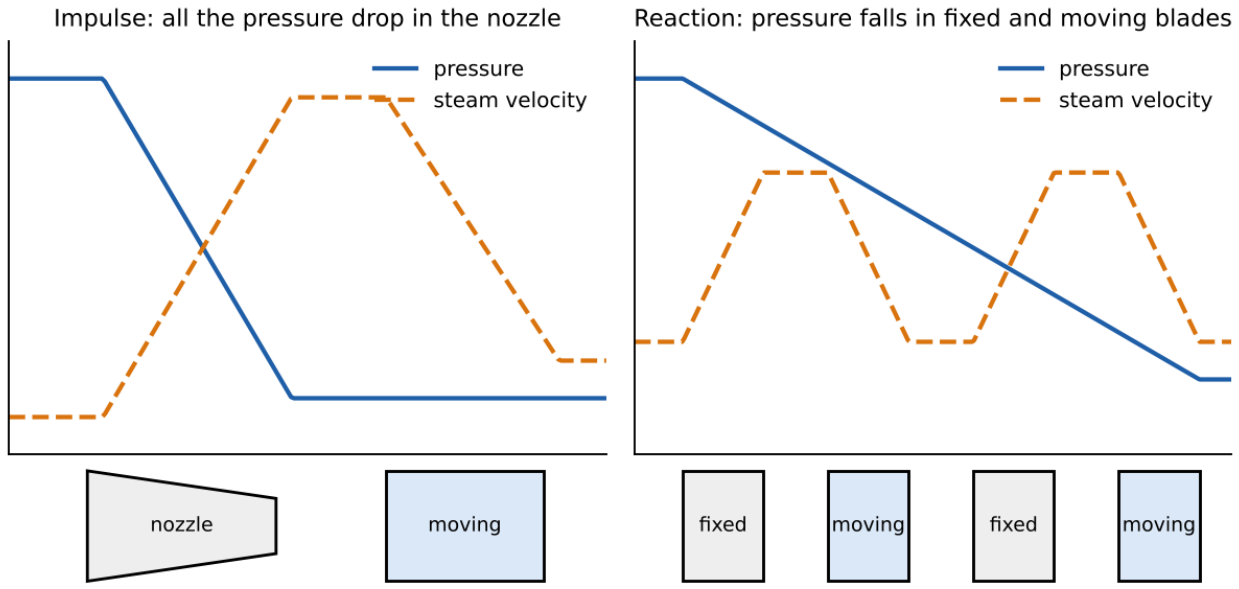


Figure 20: Pressure and velocity of the steam through an impulse stage and through reaction stages

Reading the figure. Left (impulse): the pressure (solid blue) falls only in the nozzle and stays constant across the moving blades; the velocity (dashed orange) rises in the nozzle and falls in the moving blades as the steam gives up its energy. Right (reaction): the pressure falls steadily through both the fixed and the moving blades; the velocity rises in each fixed row and falls in each moving row.

2. Answer

(i) Impulse turbine:

- The whole pressure drop of a stage takes place in **stationary nozzles**; the steam leaves them as a high-velocity jet.
- The jet strikes the **symmetrical, bucket-shaped moving blades**, which change its direction; the force on the blades comes only from this change of momentum (impulse).
- The **pressure is the same on both sides of the moving blades**, so there is no end thrust and the blade tip clearance does not cause leakage.
- A single stage (de Laval) turbine runs at very high speed, so practical turbines use **compounding**: several stages of nozzles and blades (pressure compounding, Rateau) or several rows of moving blades for one nozzle (velocity compounding, Curtis).
- Examples are the de Laval, Curtis and Rateau turbines; the Pelton wheel is the hydraulic impulse turbine.

(ii) Reaction turbine:

- The pressure drop is shared between the **fixed blades** (which act as nozzles and guide the steam) and the **moving blades** (also shaped as nozzles).

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- The moving blades are pushed both by the impulse of the steam entering them and by the **reaction** of the steam accelerating as it leaves them.
- There is a **pressure difference across the moving blades**, which causes an axial thrust (balanced by a dummy piston) and leakage over the blade tips, so the clearances must be small.
- It has many stages, each with a small pressure drop, so it runs at a moderate speed with good efficiency.
- The best-known example is the Parsons turbine (50 % reaction: equal pressure drop in the fixed and moving blades). Francis and Kaplan turbines are the hydraulic reaction turbines (4(d)).

Point	Impulse	Reaction
Pressure drop	only in the nozzles	in fixed and moving blades
Blades	symmetrical buckets	aerofoil, nozzle-shaped
Pressure across moving blades	equal	falls
Axial thrust, tip leakage	none	present
Number of stages	fewer	more

Table 5: Impulse and reaction turbines compared

6.(d) Discuss the advantages and disadvantages of a Hydro-electric power plant. Marks: 6

1. Theory

A **hydro-electric power plant** uses the energy of water stored at a height (the layout is in 5(c)). The water drives turbines (4(d)) and is returned to the river, so the "fuel" is free and is renewed by rain. Bangladesh's only hydro station is at Kaptai, on the Karnaphuli river (230 MW).

2. Answer

Advantages:

1. **No fuel cost:** water is free and renewable, so the running cost is very low.
2. **Clean:** no smoke, ash or CO_2 while running.
3. **Quick starting:** a hydro unit can start and take full load within minutes, so it is very good for **peak load** and as a reserve.
4. **Simple, reliable and long-lived:** few moving parts, low maintenance, and a life of 50 years or more; few operating staff are needed.
5. **High efficiency** (about 85–90 % from water to electricity), which does not fall with age.
6. **Multipurpose:** the dam also gives irrigation, flood control, water supply, navigation and fishing.

Disadvantages:

1. **High capital cost:** the dam, tunnels and penstocks are very costly, and construction takes a long time.
2. **Depends on rainfall:** output falls in a dry year or season.
3. **Site-bound:** suitable sites are usually in hilly areas far from the load centres, so long, costly transmission lines are needed.
4. **Environmental and social effects:** the reservoir floods land, forests and villages, people must be resettled (as at Kaptai), and river life is disturbed.
5. **Risk:** a dam failure would cause a disastrous flood.

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3. Final Concise Answer

(a) The methods are straight line, diminishing value and sinking fund. In the sinking fund method, a fixed sum $q = (P - S)\, i/[(1 + i)^n - 1]$ is set aside each year at compound interest, so that it grows to the replacement cost $P - S$ at the end of the plant's life.

(b) Coal is pulverised because the fine powder burns completely, quickly and controllably, allows any coal and very large boilers, and raises the efficiency. In a ball mill, steel balls in a slowly rotating drum crush the coal, and hot air carries the fine dust through a classifier to the burners.

(c) In an impulse turbine, all the pressure drop is in the nozzles and the jet pushes symmetrical blades at constant pressure. In a reaction turbine, the pressure falls in both fixed and moving blades, and the moving blades are driven by the reaction of the steam accelerating through them.

(d) Advantages: no fuel cost, clean, quick starting for peak load, reliable and long-lived, efficient, multipurpose. Disadvantages: high capital cost and long construction, dependence on rainfall, remote sites with long lines, and environmental, social and dam-safety risks.

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FORMULA SHEET

EEE443

Power Plant Engineering

Formula Sheet

Session 2020–21 paper

Every formula behind the 4th Year 1st Semester Final Examination, 2024

TOPICS

1 Load and plant factors — 9 formulas

2 Economic load sharing — 5 formulas

3 Boiler and economiser — 5 formulas

4 Cost of generation and tariffs — 8 formulas

5 Depreciation — 3 formulas

6 Hydro power — 1 formula

Each topic is one table: **Title** · **Equation** · **Notation** · **Reference**.

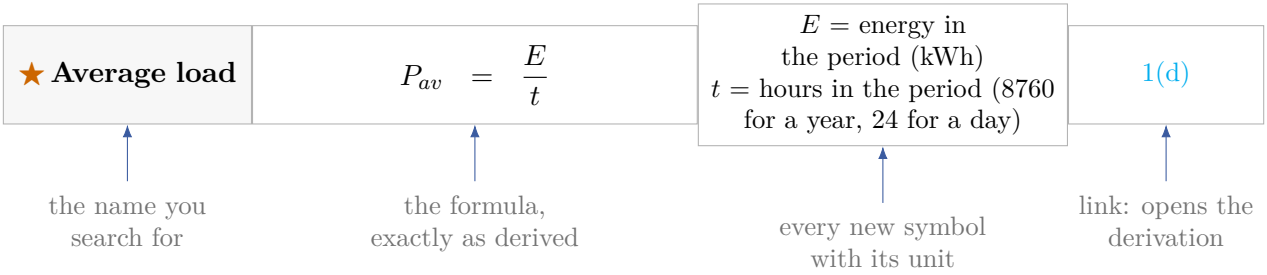
Prepared by Nur Mohammad

October 2026

BACK OF THE COVER

Before you use the sheet

How to read a row



In the exam

- Write the formula first, then put the numbers in: the formula alone often earns a mark.
- Read the symbols in the Notation column before you substitute, and check the units.
- Not sure where a formula comes from? Its Reference opens the derivation.
- The **Key formulas** page at the end gathers the rows marked ★.
- Lost a symbol? The **Symbol index** at the end links to the row where it first appears.

Contents

1 Load and plant factors 4

2 Economic load sharing 5

3 Boiler and economiser 5

4 Cost of generation and tariffs 6

5 Depreciation 6

6 Hydro power 6

7 Key formulas 7

8 Symbol index 8

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Conventions (V.K. Mehta for load, cost and tariffs; P.K. Nag for boilers and turbines): powers in kW or MW, energy in kWh or MWh; one year = 8760 h; money in taka (1 lakh = 10^5, 1 crore = 10^7, 1 Tk = 100 paisa); heat in kcal, with the specific heat of water $c_w = 1$ kcal/kg · °C, so liquid water at t °C has an enthalpy of about t kcal/kg. References point to the solutions of the 2020–21 paper.

Columns: Title · Equation · Notation · Ref. (reference).

1. Load and plant factors:

Title	Equation	Notation	Ref.
★ **Average load**	$P_{av} = \dfrac{E}{t}$	E = energy in the period (kWh) t = hours in the period (8760 for a year, 24 for a day)	1(d)
★ **Load factor**	$\text{LF} = \dfrac{P_{av}}{P_{max}}$	P_{max} = maximum demand on the station	1(d)
★ **Diversity factor**	$\text{DF} = \dfrac{\sum P_{max,i}}{P_{max}} \geq 1$	$P_{max,i}$ = maximum demand of consumer group i	1(d)
★ **Demand factor**	$\dfrac{P_{max}}{\text{connected load}} \leq 1$	connected load = total rating connected (the 30 MW plant rating in 1(d))	1(d)
★ **Capacity factor**	$\text{CF} = \dfrac{P_{av}}{C} = \dfrac{E}{C\,t}$	C = plant capacity (rated)	4(c)
★ **Plant use factor**	$\text{UF} = \dfrac{E}{C \times t_{op}}$	t_{op} = hours the plant was in operation	4(a)
Utility (utilisation) factor	$\dfrac{P_{max}}{C}$	how much of the capacity the peak used	4(a)
Reserve capacity	$C - P_{max}$	spare MW above the maximum demand	4(c)
Energy at full load as per schedule	$E_{sched} = C\,t_{op} = \dfrac{E}{\text{UF}}$	the most energy in the hours actually run	4(c)

Table 1: Load and plant factors

2. Economic load sharing:

Title	Equation	Notation	Ref.
Heat rate	$\frac{I}{P}$	I = heat input (kcal/h) P = output (MW)	2(c)
Incremental heat rate	$\frac{dI}{dP}$; in money $\frac{dF}{dP}$ (Tk/MWh)	F = fuel cost (Tk/h); extra heat or cost of one more MW	2(c)
★ **Economic sharing**	$\frac{dI_1}{dP_1} = \frac{dI_2}{dP_2} = \cdots = \lambda,\ \sum P_i = P$	λ = common incremental rate; from $dP_2 = -dP_1$ with P fixed	2(c)
Two linear units	$P_A = \frac{a_B P + b_B - b_A}{a_A + a_B}$	$dF_i/dP_i = a_i P_i + b_i$	2(d)
★ **Cost of moving load**	$\Delta F = \int_{P_1}^{P_2} \frac{dF}{dP}\, dP = \frac{a}{2}(P_2^2 - P_1^2) + b(P_2 - P_1)$	area under the incremental-cost line	2(d)

Table 2: Economic load sharing

3. Boiler and economiser:

Title	Equation	Notation	Ref.
Heat from the coal	$Q_{in} = m_f\, C$	m_f = coal burnt (kg/h) C = calorific value (kcal/kg)	3(d)
★ **Boiler efficiency**	$\eta_{boiler} = \frac{m_w(h - h_f)}{m_f\, C}$	m_w = steam (water) per hour (kg/h) h = enthalpy of the steam h_f = enthalpy of the water entering the boiler (kcal/kg)	3(d)
★ **Economiser efficiency**	$\eta_{eco} = \frac{m_w c_w (t_2 - t_1)}{m_g c_{pg}(t_g - t_a)}$	t_1, t_2 = water in and out m_g, c_{pg} = flue-gas mass and specific heat t_g = gas entering; t_a = atmosphere	3(d)
★ **Whole boiler plant**	$\eta_{plant} = \frac{m_w(h - c_w t_1)}{m_f\, C}$	water enters the economiser at t_1	3(d)
Gas temperature drop	$\Delta t_g = \frac{m_w c_w (t_2 - t_1)}{m_g c_{pg}}$	heat gained by water = heat lost by gas	3(d)

Table 3: Boiler and economiser

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4. Cost of generation and tariffs:

Title	Equation	Notation	Ref.
Energy in a year	$E = C \times \text{LF} \times 8760$	maximum demand taken as the plant capacity C (kW)	5(b)
Annual fixed charges	$(i + d)\,K + W$	K = capital cost i, d = interest and depreciation rates W = wages, taxes	5(b)
★ **Cost per kWh**	$\dfrac{\text{annual fixed charges}}{E} + \text{running cost per kWh}$	running cost = fuel, lubrication, maintenance	5(b)
Flat demand rate	$\text{bill} = a \times \text{kW}$	a = Tk per kW connected; no meter	5(a)
Straight line meter rate	$\text{bill} = b \times \text{kWh}$	b = Tk per kWh	5(a)
★ **Two-part tariff**	$\text{bill} = A \times \text{kW} + B \times \text{kWh}$	demand charge and energy charge	5(d)
Demand charge	$A = \dfrac{\text{annual fixed charges}}{\text{aggregate maximum demand of consumers}}$	Tk per kW per year	5(d)
Energy charge	$B = \dfrac{\text{annual running charges}}{(1 - \text{loss})\,E_{gen}}$	divide by the kWh **sold**, after T&D loss	5(d)

Table 4: Cost of generation and tariffs

5. Depreciation:

Title	Equation	Notation	Ref.
Straight line	$D = \dfrac{P - S}{n}$ per year	P = initial cost S = salvage value n = useful life (years)	6(a)
Diminishing value	$P(1 - r)^n = S \;\Rightarrow\; r = 1 - \left(\dfrac{S}{P}\right)^{1/n}$	r = fraction of the value left, charged each year	6(a)
★ **Sinking fund**	$q = (P - S)\,\dfrac{i}{(1 + i)^n - 1}$	q = annual deposit i = interest rate	6(a)

Table 5: Depreciation

6. Hydro power:

Title	Equation	Notation	Ref.
★ **Water power**	$P = \rho g Q H$ (W); output $= \eta\,\rho g Q H$	$\rho = 1000$ kg/m^3; $g = 9.81$ m/s^2 Q = flow (m^3/s); H = head (m) η = overall efficiency	4(d)

Table 6: Hydro power

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7. Key formulas:

The formulas marked ★ in the sheet, on one page. Each title links back to its row.

Title	Equation
Average load	$P_{av} = \dfrac{E}{t}$
Load factor	$\text{LF} = \dfrac{P_{av}}{P_{max}}$
Diversity factor	$\text{DF} = \dfrac{\sum P_{max,i}}{P_{max}} \geq 1$
Demand factor	$\dfrac{P_{max}}{\text{connected load}} \leq 1$
Capacity factor	$\text{CF} = \dfrac{P_{av}}{C} = \dfrac{E}{C\,t}$
Plant use factor	$\text{UF} = \dfrac{E}{C \times t_{op}}$
Economic sharing	$\dfrac{dI_1}{dP_1} = \dfrac{dI_2}{dP_2} = \cdots = \lambda,\ \sum P_i = P$
Cost of moving load	$\Delta F = \displaystyle\int_{P_1}^{P_2} \frac{dF}{dP}\,dP = \tfrac{a}{2}(P_2^2 - P_1^2) + b(P_2 - P_1)$
Boiler efficiency	$\eta_{boiler} = \dfrac{m_w(h - h_f)}{m_f\,C}$
Economiser efficiency	$\eta_{eco} = \dfrac{m_w c_w (t_2 - t_1)}{m_g c_{pg} (t_g - t_a)}$
Whole boiler plant	$\eta_{plant} = \dfrac{m_w(h - c_w t_1)}{m_f\,C}$
Cost per kWh	$\dfrac{\text{annual fixed charges}}{E} + \text{running cost per kWh}$
Two-part tariff	$\text{bill} = A \times \text{kW} + B \times \text{kWh}$
Sinking fund	$q = (P - S)\,\dfrac{i}{(1+i)^n - 1}$
Water power	$P = \rho g Q H$ (W); output $= \eta\,\rho g Q H$

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8. Symbol index:

Every symbol the sheet defines, in the order it first appears. The last column links to that row.

Symbol	Meaning	First used in
E	energy in the period (kWh)	Average load
t	hours in the period (8760 for a year, 24 for a day)	Average load
P_{max}	maximum demand on the station	Load factor
$P_{max,i}$	maximum demand of consumer group i	Diversity factor
connected load	total rating connected (the 30 MW plant rating in 1(d))	Demand factor
C	plant capacity (rated)	Capacity factor
t_{op}	hours the plant was in operation	Plant use factor
I	heat input (kcal/h)	Heat rate
P	output (MW)	Heat rate
F	fuel cost (Tk/h); extra heat or cost of one more MW	Incremental heat rate
λ	common incremental rate; from $dP_2 = -dP_1$ with P fixed	Economic sharing
m_f	coal burnt (kg/h)	Heat from the coal
m_w	steam (water) per hour (kg/h)	Boiler efficiency
h	enthalpy of the steam	Boiler efficiency
h_f	enthalpy of the water entering the boiler (kcal/kg)	Boiler efficiency
t_1, t_2	water in and out	Economiser efficiency
m_g, c_{pg}	flue-gas mass and specific heat	Economiser efficiency
t_g	gas entering; t_a = atmosphere	Economiser efficiency
heat gained by water	heat lost by gas	Gas temperature drop
K	capital cost	Annual fixed charges
i, d	interest and depreciation rates	Annual fixed charges
W	wages, taxes	Annual fixed charges
running cost	fuel, lubrication, maintenance	Cost per kWh
a	Tk per kW connected; no meter	Flat demand rate
b	Tk per kWh	Straight line meter rate
S	salvage value	Straight line
n	useful life (years)	Straight line
r	fraction of the value left, charged each year	Diminishing value
q	annual deposit	Sinking fund
i	interest rate	Sinking fund
ρ	1000 kg/m^3; g = 9.81 m/s^2	Water power
Q	flow (m^3/s); H = head (m)	Water power
η	overall efficiency	Water power

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